Text as Data

W7: Scaling Political Texts

Yen-Chieh Liao 廖彥傑

2026-10-23

Fall 2026 (115-1) · National Taiwan University (GIND)
NTLDEV 7191 · Fri. 16:30–18:20 · 國發 204
Last compiled: 2026-10-30 12:33:35 (Asia/Taipei)



National Taiwan University
國立臺灣大學

Calls for papers: 2027

- IPSA World Congress 2027 · Rome
  Panel: Special Interest Influence and the Quality of Democratic Representation
  Abstract deadline: 4 November 2026
- EPSS 2027 Conference · Copenhagen
  Paper deadline: 13 November 2026, 23:59 CET
  Taipei time: **14 November 2026, 06:59**
- 22nd TASPAA Annual Conference · 2027
  Abstract and panel proposal deadline: 1 December 2026
  14–15 May 2027 · National Taipei University, Sanxia Campus

Official calls: IPSA · EPSS · TASPAA.

This weekend in Taipei

Saturday, 31 October 2026

Development Studies Conference
第十八屆發展研究年會



NTU · GIND（國發所）
Keynote 10:40–12:00 · Room 302

Taipei Pride
第24屆臺灣同志遊行



Taipei City Hall Plaza
Gather from 13:00 · Parade starts at 14:05

Have a good weekend. Enjoy Taiwan’s diversity and inclusive spirit.

Official event information and images: Development Studies Conference · Taiwan LGBT+ Pride.

Parliamentary origins: Ireland, 1991

An early Wordscores application: Laver & Benoit (2002).

Research choice	The Irish Dáil study
Political setting	**October 1991 confidence debate on the Fianna Fáil–PD government**
Texts	58 English-language speeches by individual legislators
Measurement target	How strongly did each speaker support or oppose the government?

Discuss: what might speeches reveal beyond a final confidence vote?

Laver & Benoit (2002), *Irish Political Studies* 17(1): 59–73. DOI · Original article.

Parliamentary origins: Ireland, 1991

An early Wordscores application: Laver & Benoit (2002).

Research choice	The Irish Dáil study
Political setting	October 1991 confidence debate on the Fianna Fáil–PD government
Texts	**58 English-language speeches by individual legislators**
Measurement target	How strongly did each speaker support or oppose the government?

Discuss: what might speeches reveal beyond a final confidence vote?

Laver & Benoit (2002), *Irish Political Studies* 17(1): 59–73. DOI · Original article.

Parliamentary origins: Ireland, 1991

An early Wordscores application: Laver & Benoit (2002).

Research choice	The Irish Dáil study
Political setting	October 1991 confidence debate on the Fianna Fáil–PD government
Texts	58 English-language speeches by individual legislators
Measurement target	**How strongly did each speaker support or oppose the government?**

Discuss: what might speeches reveal beyond a final confidence vote?

Laver & Benoit (2002), *Irish Political Studies* 17(1): 59–73. DOI · Original article.

From reference speeches to positions

Step	The early parliamentary design
Assign reference positions	Haughey: **+1**. Bruton and de Rossa: **−1**
Score the vocabulary	Learn word scores from those reference speeches
Locate other speakers	Combine the scored words in each remaining speech

Interpretation: support for this government in this debate.

2002: parliamentary application.
2003: the general Wordscores article, including this debate.

Laver & Benoit (2002), pp. 61–66, reference speeches. Laver, Benoit & Garry (2003), pp. 326–328, APSR.

Text labels and continuous positions

Research task	W6 connection	W7 question
Assign a category	Hate speech and grandstanding	Which texts belong together?
Measure a degree	Julia Park's original grandstanding score (continuous)	Where does a text sit on a scale?
Interpret the measure	Read errors and external evidence	What gives a position its meaning?

Today: Wordscores, Wordfish, Wordshoal and Latent Semantic Scaling (LSS).

Course connection: W6: Human Coding and Document Classification. Original continuous measure: Park (2021). Lecture context: Benoit (2018), pp. 3–4, 21–27.

Text labels and continuous positions

Research task	W6 connection	W7 question
Assign a category	**Hate speech and grandstanding**	Which texts belong together?
Measure a degree	**Julia Park's original grandstanding score (continuous)**	Where does a text sit on a scale?
Interpret the measure	**Read errors and external evidence**	What gives a position its meaning?

Today: Wordscores, Wordfish, Wordshoal and Latent Semantic Scaling (LSS).

Course connection: W6: Human Coding and Document Classification. Original continuous measure: Park (2021). Lecture context: Benoit (2018), pp. 3–4, 21–27.

Text labels and continuous positions

Research task	W6 connection	W7 question
Assign a category	Hate speech and grandstanding	**Which texts belong together?**
Measure a degree	Julia Park's original grandstanding score (continuous)	**Where does a text sit on a scale?**
Interpret the measure	Read errors and external evidence	**What gives a position its meaning?**

Today: Wordscores, Wordfish, Wordshoal and Latent Semantic Scaling (LSS).

Course connection: W6: Human Coding and Document Classification. Original continuous measure: Park (2021). Lecture context: Benoit (2018), pp. 3–4, 21–27.

Learning objectives

- Explain the problem each method addresses and its main assumptions.
- Connect a published application to its texts and measurement target.
- Distinguish reference scores, latent positions and seed words.
- Read the texts and plan independent checks before fitting a model.

Sequence: cases → method concepts
→ reading and validation → R tools and practice.

Research cases and original-study sources follow. Method selection depends on the question and available evidence.

Research cases in W7

Study	Article	Method	Texts	Teaching connection
Study 1	Laver, Benoit & Garry (2003)	Wordscores	Party manifestos	Wordscores and reference positions
Study 2	Slapin & Proksch (2008)	Wordfish	German manifestos	Wordfish and change over time
Study 6	Lowe & Benoit (2013)	Wordfish	Irish budget speeches	Human judgments and model failure
Study 3	Lauderdale & Herzog (2016)	Wordshoal	Legislative debates	Wordshoal and debate structure
Study 4	Trubowitz & Watanabe (2021)	LSS	News summaries	LSS and geopolitical threat
Study 5	Hjorth et al. (2015)	Both	Danish and German manifestos	External benchmarks and context

Both = Wordscores + Wordfish. Wordshoal uses Wordfish within each debate.

Wordscores · Wordfish · Wordshoal · GTI · External benchmarks · Human validation.

Wordscores: reference texts and new documents

Situation: some texts have credible positions on a defined dimension.

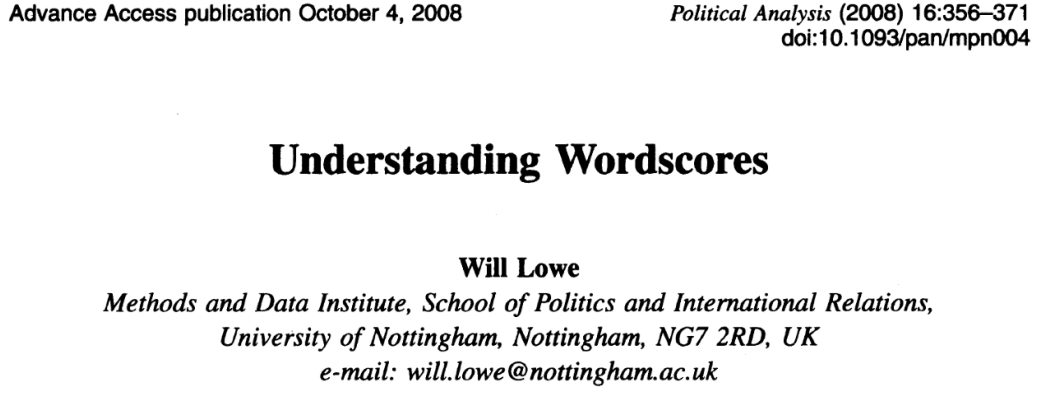
- Use these documents as **reference texts**.
- Learn how their words distinguish those positions.
- Estimate positions for comparable new documents.

Example: earlier party manifestos provide references for a later election.

Laver, Benoit & Garry (2003), APSR, pp. 312–314. Lecture context: Benoit (2018), pp. 4, 17–20.

Wordscores: assumptions and scope

- **Reference scores:** measure the concept we want.
- **Comparable language:** words signal positions similarly.
- **Coverage:** new texts contain enough scored words.
- **Transfer:** word–position links carry over to new texts.



Advance Access publication October 4, 2008

Political Analysis (2008) 16:356–371
doi:10.1093/pan/mpn004

Understanding Wordscores

Will Lowe
Methods and Data Institute, School of Politics and International Relations, University of Nottingham, Nottingham, NG7 2RD, UK
e-mail: will.lowe@nottingham.ac.uk

Will Lowe (2008)

Political Analysis 16(4): 356–371

Discuss: would a campaign manifesto be a suitable reference for a parliamentary interruption?

Model interpretation: Lowe (2008), *Understanding Wordscores*. Course checks: Laver, Benoit & Garry (2003), pp. 314–315; Benoit (2018), pp. 18–20. Image: user-supplied article screenshot. Discussion prompt is a course application.

Laver, Benoit and Garry: party positions

Study 1

American Political Science Review Vol. 97, No. 2 May 2003

Extracting Policy Positions from Political Texts Using Words as Data

MICHAEL LAVER and KENNETH BENOIT *Trinity College, University of Dublin*
JOHN GARRY *University of Reading*

Michael Laver, Kenneth Benoit and John Garry

American Political Science Review, 2003

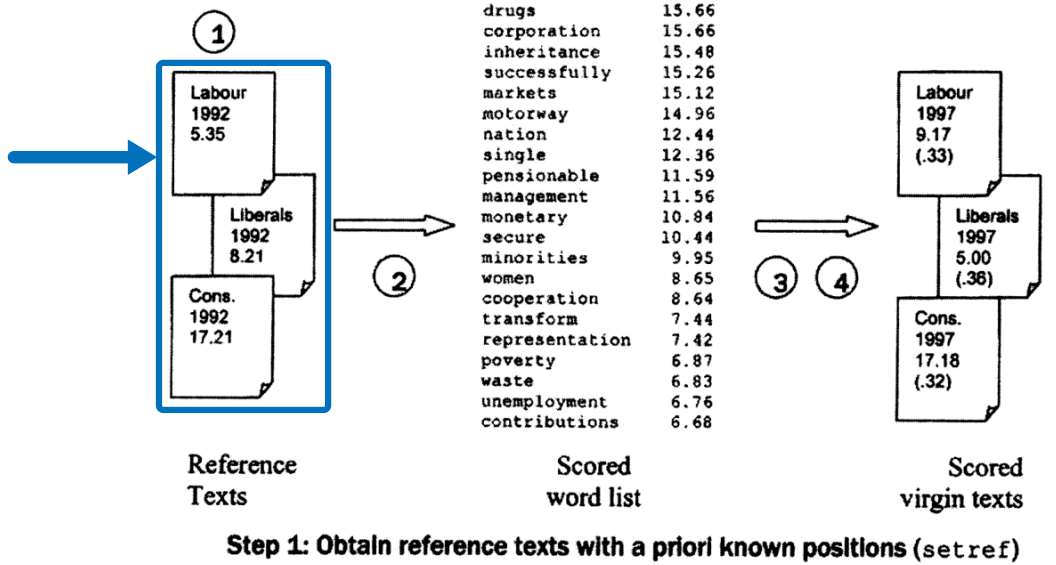
- **Question:** how can texts locate parties on policy dimensions?
- **References:** British 1992 manifestos and expert scores.
- **New texts:** the parties' 1997 manifestos.

Laver, Benoit & Garry (2003), "Extracting Policy Positions from Political Texts Using Words as Data". Lecture walkthrough: Benoit (2018), pp. 5–8.

Wordscores: interpreting estimated change

Study 1

FIGURE 1. The Wordscore procedure, using the British 1992–1997 manifesto scoring as an illustration



Step 1: Obtain reference texts with a priori known positions (`setref`)
Step 2: Generate word scores from reference texts (`wordscore`)
Step 3: Score each virgin text using word scores (`textscore`)
Step 4: (optional) Transform virgin text scores to original metric

Note: Scores for 1997 virgin texts are transformed estimated scores; parenthetical values are standard errors. The scored word list is a sample of the 5,299 total words scored from the three reference texts.

Figure 1 · reference texts → words → new texts

1 · 1992 references

- Labour: **5.35**
- Liberal Democrats: **8.21**
- Conservatives: **17.21**

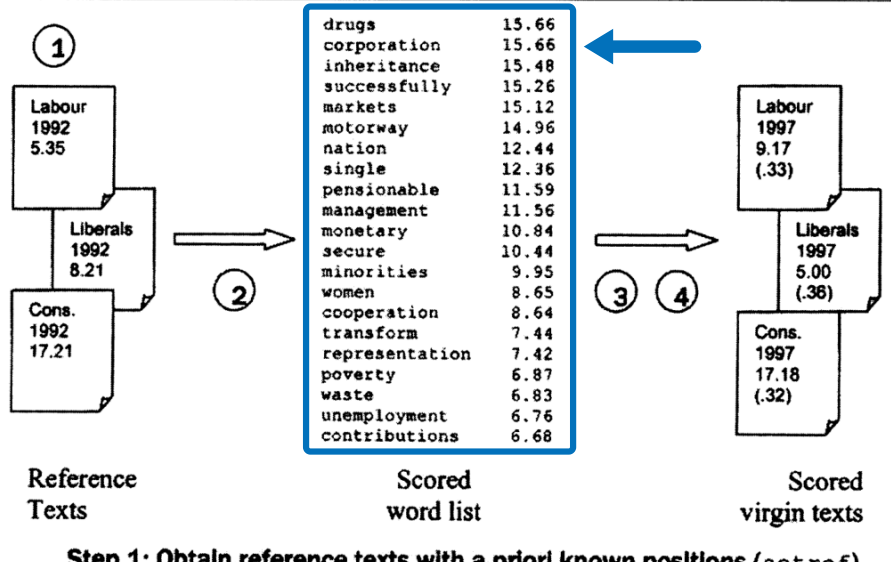
Economic policy:
lower = public services
higher = tax cuts

Laver, Benoit & Garry (2003), APSR, Figure 1, p. 314. Original figure, including its transformation note. Lecture walkthrough: Benoit (2018), pp. 5–8, 15. Blue arrows and outlines are course annotations.

Wordscores: interpreting estimated change

Study 1

FIGURE 1. The Wordscore procedure, using the British 1992–1997 manifesto scoring as an illustration



Step 1: Obtain reference texts with a priori known positions (`setref`)
Step 2: Generate word scores from reference texts (`wordscore`)
Step 3: Score each virgin text using word scores (`textscore`)
Step 4: (optional) Transform virgin text scores to original metric

Note: Scores for 1997 virgin texts are transformed estimated scores; parenthetical values are standard errors. The scored word list is a sample of the 5,299 total words scored from the three reference texts.

Figure 1 · reference texts → words → new texts

2 · From texts to words

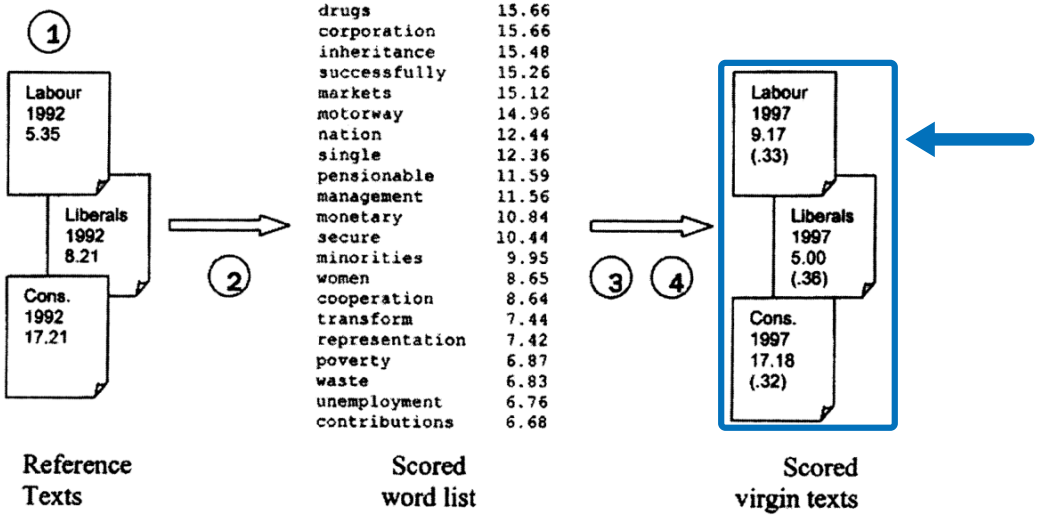
- Learn word scores from the 1992 references.
- Combine them using word proportions in each 1997 manifesto.

Laver, Benoit & Garry (2003), APSR, Figure 1, p. 314. Original figure, including its transformation note. Lecture walkthrough: Benoit (2018), pp. 5–8, 15. Blue arrows and outlines are course annotations.

Wordscores: interpreting estimated change

Study 1

FIGURE 1. The Wordscore procedure, using the British 1992–1997 manifesto scoring as an illustration



Step 1: Obtain reference texts with a priori known positions (`setref`)
Step 2: Generate word scores from reference texts (`wordscore`)
Step 3: Score each virgin text using word scores (`textscore`)
Step 4: (optional) Transform virgin text scores to original metric

Note: Scores for 1997 virgin texts are transformed estimated scores; parenthetical values are standard errors. The scored word list is a sample of the 5,299 total words scored from the three reference texts.

Figure 1 · reference texts → words → new texts

3 · 1997 positions

- Labour: **9.17**
- Liberal Democrats: **5.00**
- Conservatives: **17.18**

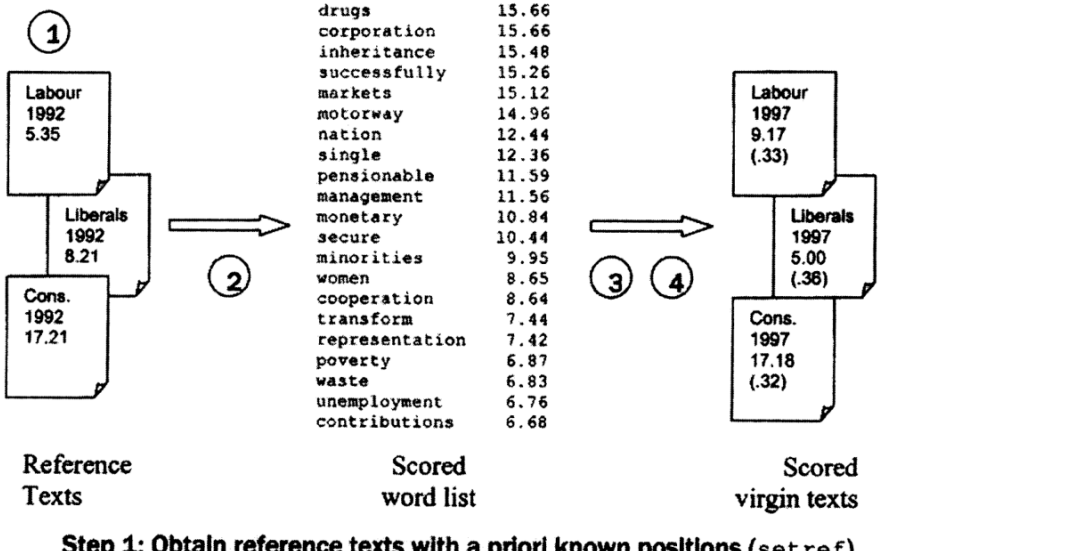
Labour and the Liberal Democrats switch order.

Laver, Benoit & Garry (2003), APSR, Figure 1, p. 314. Original figure, including its transformation note. Lecture walkthrough: Benoit (2018), pp. 5–8, 15. Blue arrows and outlines are course annotations.

Wordscores: interpreting estimated change

Study 1

FIGURE 1. The Wordscore procedure, using the British 1992–1997 manifesto scoring as an illustration



Note: Scores for 1997 virgin texts are transformed estimated scores; parenthetical values are standard errors. The scored word list is a sample of the 5,299 total words scored from the three reference texts.

Figure 1 · reference texts → words → new texts

4 · Read the note

- Values are **transformed**.
- Parentheses are **standard errors**.
- Compare ordering; be cautious about distances.

Laver, Benoit & Garry (2003), APSR, Figure 1, p. 314. Original figure, including its transformation note. Lecture walkthrough: Benoit (2018), pp. 5–8, 15. Blue arrows and outlines are course annotations.

Wordfish: positions without reference scores

Situation: we have comparable documents but no trusted position labels.

- Estimate a latent dimension from patterns in word counts.
- Account for document length and the baseline frequency of words.
- Interpret the recovered dimension using texts and external evidence.

Example: party manifestos from several elections, or speeches in one budget debate.

Slapin & Proksch (2008), AJPS. Lowe & Benoit (2013), Political Analysis. Lecture context: Benoit (2018), pp. 34–35, 37.

Wordfish: assumptions about word counts

- **Count model: word counts follow conditionally independent Poisson distributions.**
- **One dimension:** a document position helps explain word use after accounting for document and word baselines.
- **Political interpretation:** the dimension must track the concept we care about.

Possible complications: changing topics, speaking roles, genre and historical vocabulary.

Slapin & Proksch (2008), model specification, pp. 708–710. Interpretation also follows Lowe & Benoit (2013). Lecture context: Benoit (2018), pp. 37–38, 42–43.

Wordfish: assumptions about word counts

- **Count model:** word counts follow conditionally independent Poisson distributions.
- **One dimension: a document position helps explain word use after accounting for document and word baselines.**
- **Political interpretation:** the dimension must track the concept we care about.

Possible complications: changing topics, speaking roles, genre and historical vocabulary.

Slapin & Proksch (2008), model specification, pp. 708–710. Interpretation also follows Lowe & Benoit (2013). Lecture context: Benoit (2018), pp. 37–38, 42–43.

Wordfish: assumptions about word counts

- **Count model:** word counts follow conditionally independent Poisson distributions.
- **One dimension:** a document position helps explain word use after accounting for document and word baselines.
- **Political interpretation: the dimension must track the concept we care about.**

Possible complications: changing topics, speaking roles, genre and historical vocabulary.

Slapin & Proksch (2008), model specification, pp. 708–710. Interpretation also follows Lowe & Benoit (2013). Lecture context: Benoit (2018), pp. 37–38, 42–43.

Slapin and Proksch: German manifestos

Study 2

A Scaling Model for Estimating Time-Series Party Positions from Texts

Jonathan B. Slapin Trinity College, Dublin
Sven-Oliver Proksch University of California, Los Angeles

Jonathan B. Slapin and Sven-Oliver Proksch

American Journal of Political Science, 2008

1 · Model the word counts

$$y_{ij} \sim \text{Poisson}(\mu_{ij})$$

$$\log \mu_{ij} = \alpha_i + \psi_j + \beta_j \theta_i$$

- y_{ij}: observed word count.
- μ_{ij}: expected word count.
- No reference scores supplied.

Slapin & Proksch (2008), AJPS, p. 709. Course notation: i indexes a manifesto, j a word. Lecture context: Benoit (2018), pp. 37–38. Red circles and arrows are course annotations.

Slapin and Proksch: German manifestos

Study 2

A Scaling Model for Estimating Time-Series Party Positions from Texts

Jonathan B. Slapin Trinity College, Dublin
Sven-Oliver Proksch University of California, Los Angeles

Jonathan B. Slapin and Sven-Oliver Proksch

American Journal of Political Science, 2008

2 · Allow different baselines

$$y_{ij} \sim \text{Poisson}(\mu_{ij})$$

$$\log \mu_{ij} = \alpha_i + \psi_j + \beta_j \theta_i$$

- α_i: document baseline (partly length).
- ψ_j: word baseline (overall frequency).

Slapin & Proksch (2008), AJPS, p. 709. Course notation: i indexes a manifesto, j a word. Lecture context: Benoit (2018), pp. 37–38. Red circles and arrows are course annotations.

Slapin and Proksch: German manifestos

Study 2

A Scaling Model for Estimating Time-Series Party Positions from Texts

Jonathan B. Slapin Trinity College, Dublin
Sven-Oliver Proksch University of California, Los Angeles

Jonathan B. Slapin and Sven-Oliver Proksch

American Journal of Political Science, 2008

3 · Connect words and positions

$$y_{ij} \sim \text{Poisson}(\mu_{ij})$$

$$\log \mu_{ij} = \alpha_i + \psi_j + \beta_j \theta_i$$

- θ_i: manifesto position.
- β_j: word loading (log scale).
- Fit jointly across elections.

Slapin & Proksch (2008), AJPS, p. 709. Course notation: i indexes a manifesto, j a word. Lecture context: Benoit (2018), pp. 37–38. Red circles and arrows are course annotations.

Slapin and Proksch: German manifestos

Study 2

A Scaling Model for Estimating Time-Series Party Positions from Texts

Jonathan B. Slapin Trinity College, Dublin
Sven-Oliver Proksch University of California, Los Angeles

Jonathan B. Slapin and Sven-Oliver Proksch

American Journal of Political Science, 2008

- **Question:** how do party positions change over time?
- **Texts:** German manifestos, 1990–2005.
- **Evidence:** estimated positions and politically discriminating words.

Slapin & Proksch (2008), "A Scaling Model for Estimating Time-Series Party Positions from Texts". Lecture example: Benoit (2018), p. 46.

German party positions across elections

Study 2

FIGURE 1 Estimated Party Positions in Germany, 1990–2005

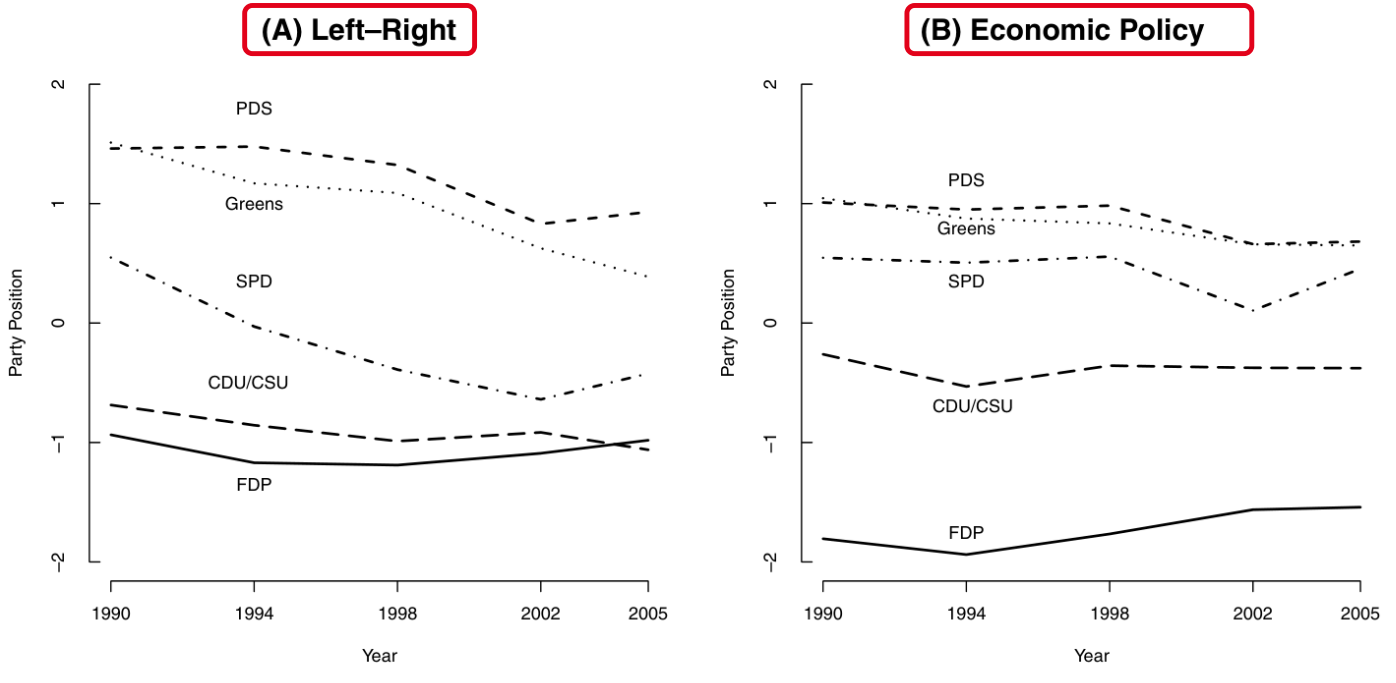


Figure 1 · panels A and B · course annotations in red

Panel A vs. panel B

- Compare parties **within each panel.**
- **Panels use different policy scopes.**
- A common time scale requires comparable word use.

Slapin & Proksch (2008), AJPS, Figure 1, p. 714. Original panels A–B, cropped from the four-panel figure. Lecture example: Benoit (2018), p. 46.

German party positions across elections

Study 2

FIGURE 1 Estimated Party Positions in Germany, 1990–2005

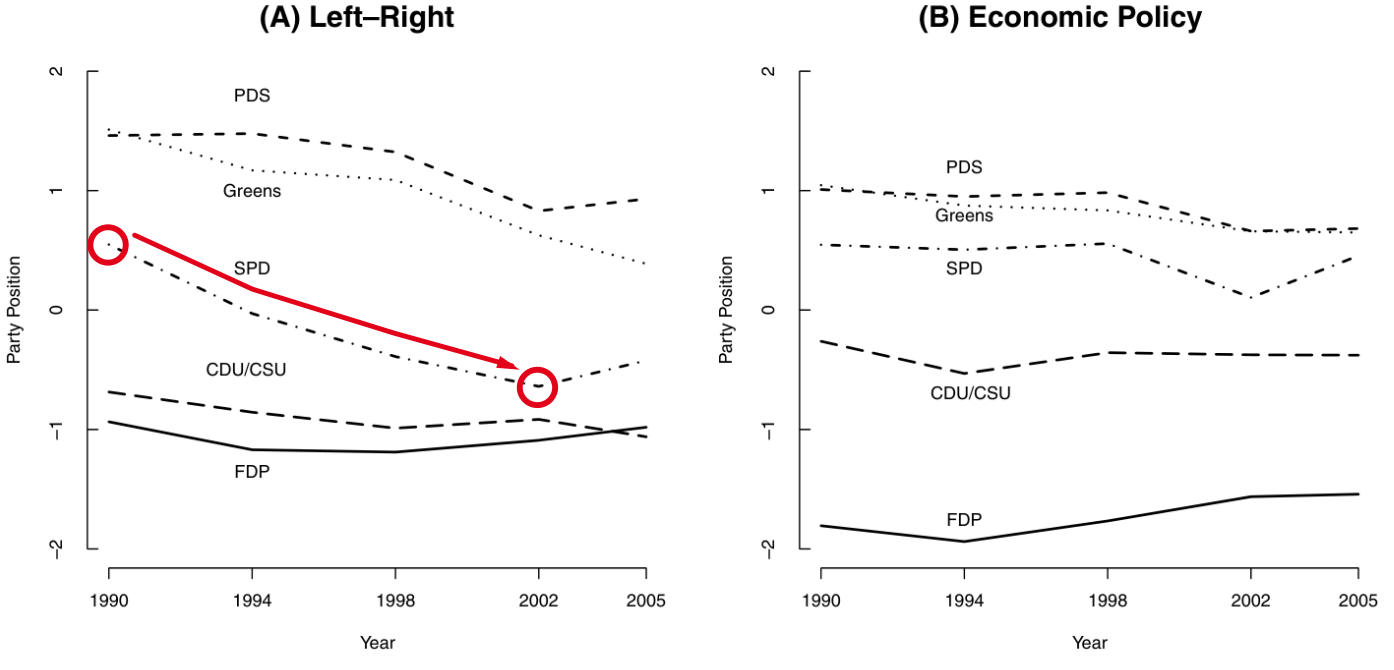


Figure 1 · panels A and B · course annotations in red

SPD: 1990–2002

- **Compare parties within each panel.**
- Panels use different policy scopes.
- A common time scale requires comparable word use.

Slapin & Proksch (2008), AJPS, Figure 1, p. 714. Original panels A–B, cropped from the four-panel figure. Lecture example: Benoit (2018), p. 46.

Wordfish: reading the word plot

Study 2 · word parameters

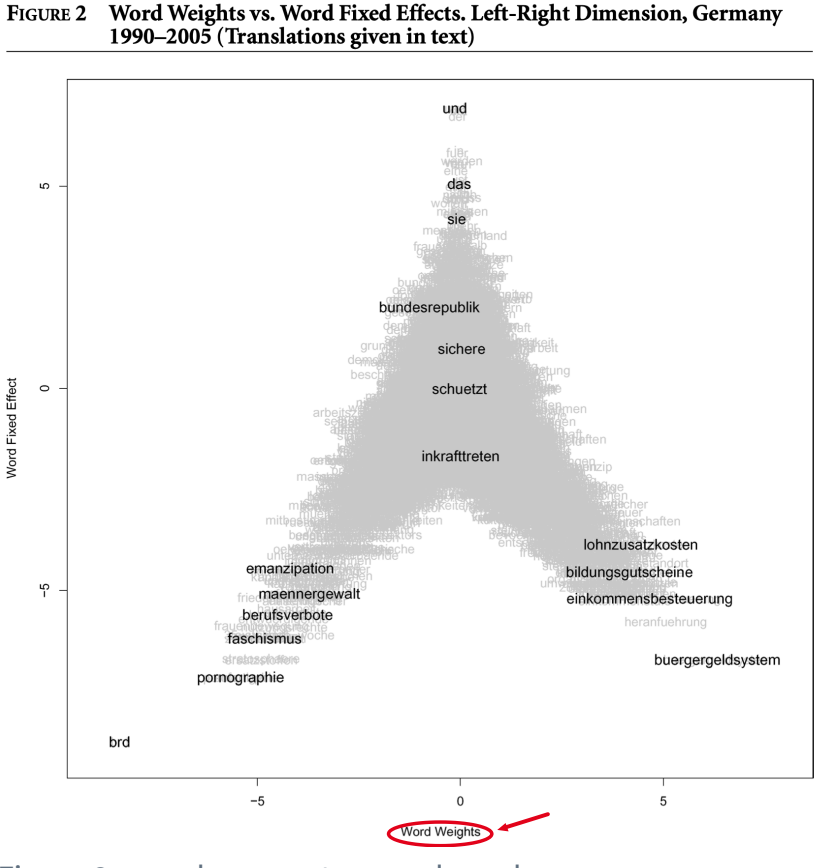


Figure 2 · word parameters · red overlays

Read across, then up.

- **x: word weight (β). Farther from 0 means more separation.**
- **y: word fixed effect (ψ).** Higher means more common baseline use.
- **Top centre:** *und* (“and”) is common but has little weight.
- **Lower sides:** read the distinctive words in context.

Slapin & Proksch (2008), AJPS, Figure 2, p. 715. Original image supplied by the instructor.

Wordfish: reading the word plot

Study 2 · word parameters

FIGURE 2 **Word Weights vs. Word Fixed Effects. Left-Right Dimension, Germany 1990–2005 (Translations given in text)**

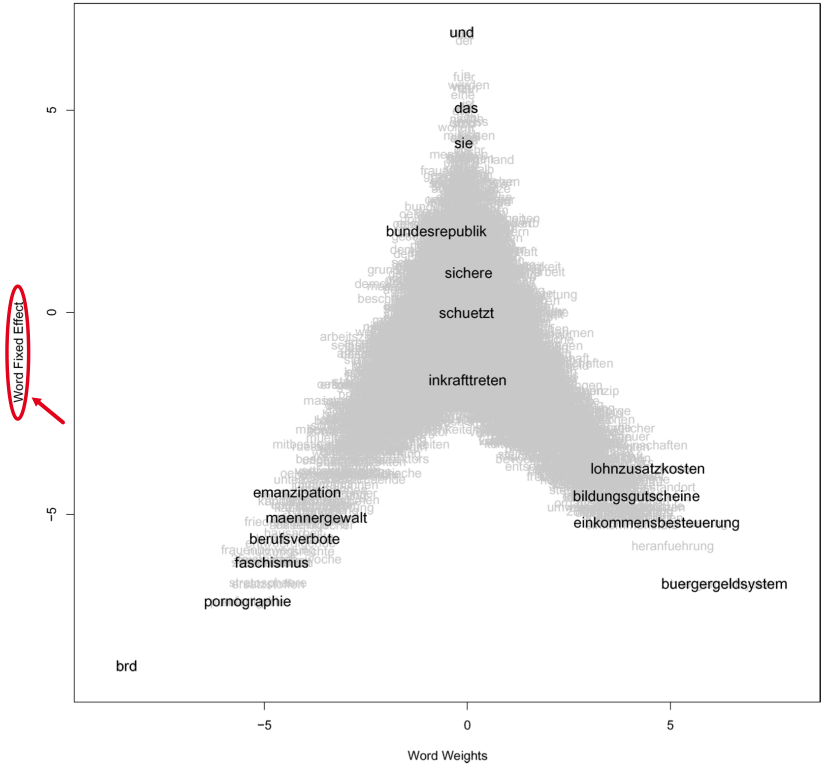


Figure 2 · word parameters · red overlays

Read across, then up.

- **x: word weight (β).** Farther from 0 means more separation.
- **y: word fixed effect (ψ). Higher means more common baseline use.**
- **Top centre:** *und* (“and”) is common but has little weight.
- **Lower sides:** read the distinctive words in context.

Slapin & Proksch (2008), AJPS, Figure 2, p. 715. Original image supplied by the instructor.

Wordfish: reading the word plot

Study 2 · word parameters

FIGURE 2 **Word Weights vs. Word Fixed Effects. Left-Right Dimension, Germany 1990–2005 (Translations given in text)**

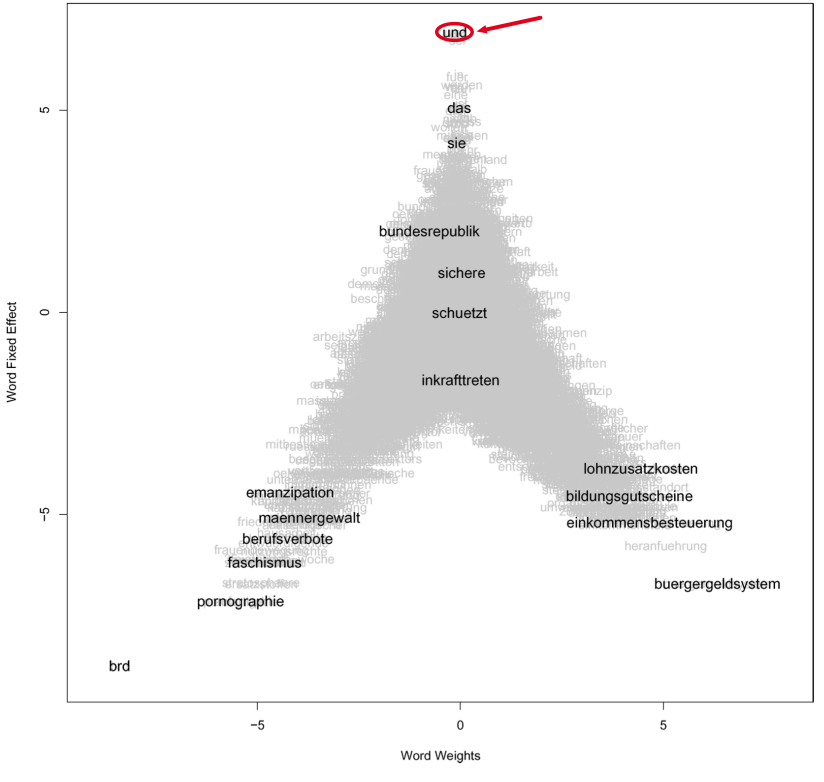


Figure 2 · word parameters · red overlays

Read across, then up.

- **x: word weight (β).** Farther from 0 means more separation.
- **y: word fixed effect (ψ).** Higher means more common baseline use.
- **Top centre: *und* (“and”) is common but has little weight.**
- **Lower sides:** read the distinctive words in context.

Slapin & Proksch (2008), AJPS, Figure 2, p. 715. Original image supplied by the instructor.

Wordfish: reading the word plot

Study 2 · word parameters

FIGURE 2 **Word Weights vs. Word Fixed Effects. Left-Right Dimension, Germany 1990–2005 (Translations given in text)**

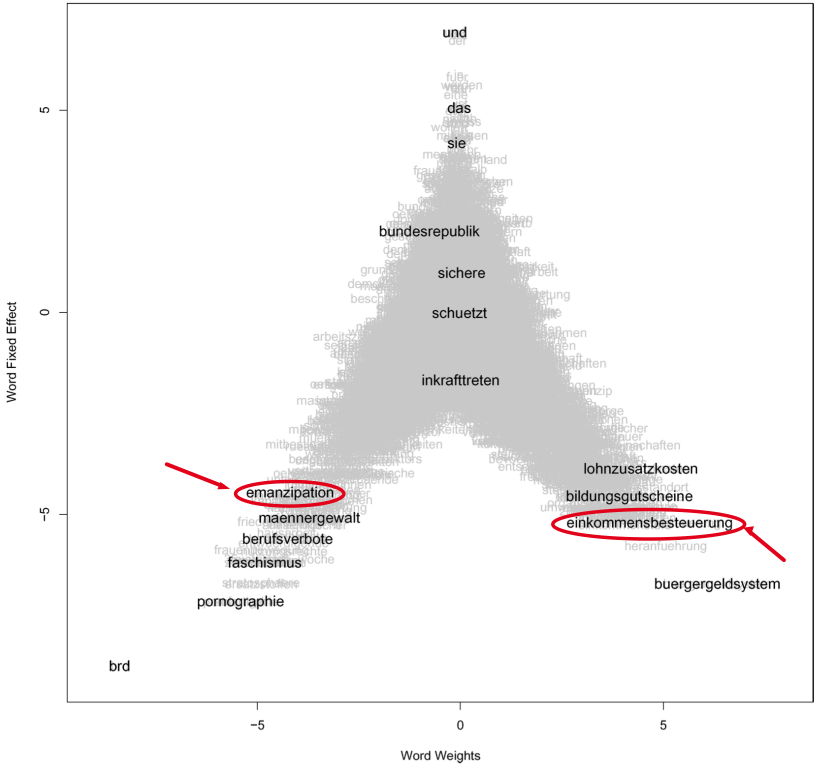


Figure 2 · word parameters · red overlays

Read across, then up.

- **x: word weight (β).** Farther from 0 means more separation.
- **y: word fixed effect (ψ).** Higher means more common baseline use.
- **Top centre:** *und* (“and”) is common but has little weight.
- **Lower sides: read the distinctive words in context.**

Slapin & Proksch (2008), AJPS, Figure 2, p. 715. Original image supplied by the instructor.

Wordscores and Wordfish: strengths and limits

Aspect	Wordscores	Wordfish
Strength	**Scores new texts using reference positions.**	Estimates positions without supplied position scores.
Assumptions	**Trusted reference positions; comparable language; shared word–position links.**	One common dimension; conditionally independent Poisson counts; shared word effects.
Limits	**Reference choice matters; unseen words lack scores; shared words pull raw scores inward.**	Topics or roles may drive the scale; small standard errors do not establish validity.

Both need checks against the texts and independent evidence.

Reference: Laver et al. (2003), Slapin & Proksch (2008), Lowe & Benoit (2013). Benoit (2018), pp. 18–20, 37–45, 59.

Wordscores and Wordfish: strengths and limits

Aspect	Wordscores	Wordfish
Strength	Scores new texts using reference positions.	**Estimates positions without supplied position scores.**
Assumptions	Trusted reference positions; comparable language; shared word–position links.	**One common dimension; conditionally independent Poisson counts; shared word effects.**
Limits	Reference choice matters; unseen words lack scores; shared words pull raw scores inward.	**Topics or roles may drive the scale; small standard errors do not establish validity.**

Both need checks against the texts and independent evidence.

Reference: Laver et al. (2003), Slapin & Proksch (2008), Lowe & Benoit (2013). Benoit (2018), pp. 18–20, 37–45, 59.

Wordscores and Wordfish: strengths and limits

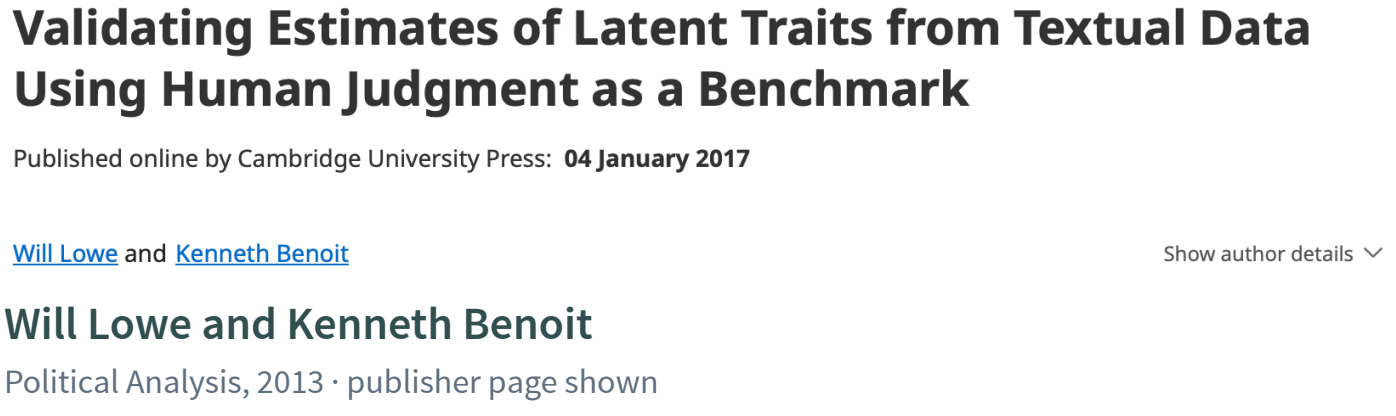
Aspect	Wordscores	Wordfish
Strength	Scores new texts using reference positions.	Estimates positions without supplied position scores.
Assumptions	Trusted reference positions; comparable language; shared word–position links.	One common dimension; conditionally independent Poisson counts; shared word effects.
Limits	**Reference choice matters; unseen words lack scores; shared words pull raw scores inward.**	**Topics or roles may drive the scale; small standard errors do not establish validity.**

Both need checks against the texts and independent evidence.

Reference: Laver et al. (2003), Slapin & Proksch (2008), Lowe & Benoit (2013). Benoit (2018), pp. 18–20, 37–45, 59.

Lowe and Benoit: the Irish budget debate

Study 6



Validating Estimates of Latent Traits from Textual Data Using Human Judgment as a Benchmark

Published online by Cambridge University Press: **04 January 2017**

Will Lowe and Kenneth Benoit

Show author details

Will Lowe and Kenneth Benoit

Political Analysis, 2013 · publisher page shown

- **Texts:** 14 speeches on the 2010 Irish budget.
- **Debate:** December 2009.
- **Benchmark:** human judgments of budget support.

Lowe & Benoit (2013), Political Analysis, §3. Image: publisher page (user-provided screenshot).

Human judgments expose a model failure

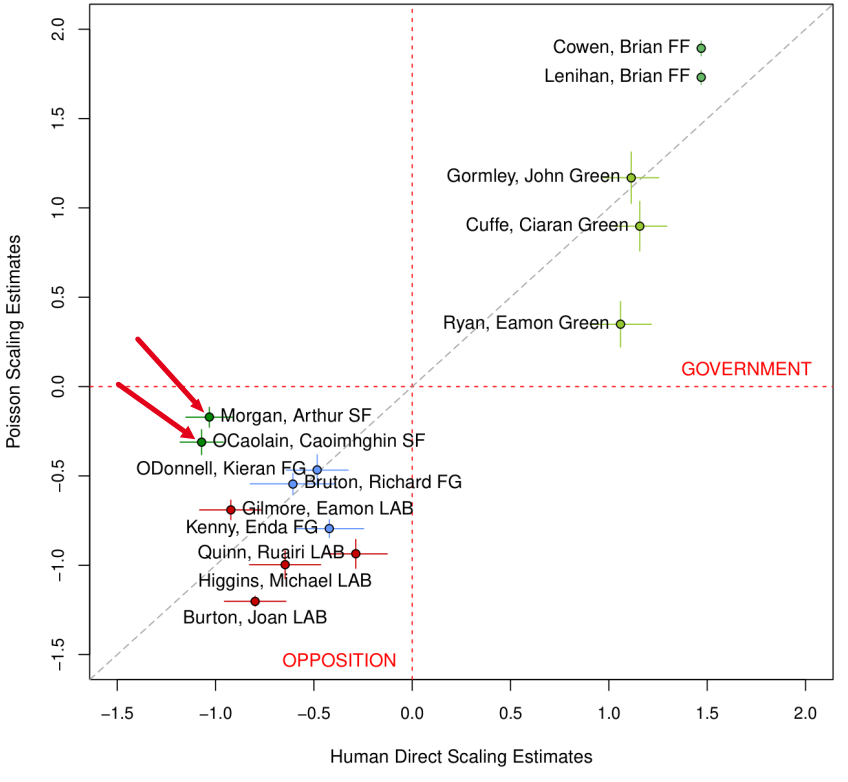


Figure 3: *Direct comparison of Human v. Wordfish Estimates.*

Figure 3 · red arrows identify Sinn Féin

Study 6

- Many positions broadly agree.
- **Sinn Féin: strong opposition looks too moderate in Wordfish.**
- A second political distinction complicates one dimension.

Lowe & Benoit (2013), Political Analysis, §§5–6. Figure 3 in the author manuscript, p. 32. Dimensionality background: Benoit (2018), p. 63.

Human judgments expose a model failure

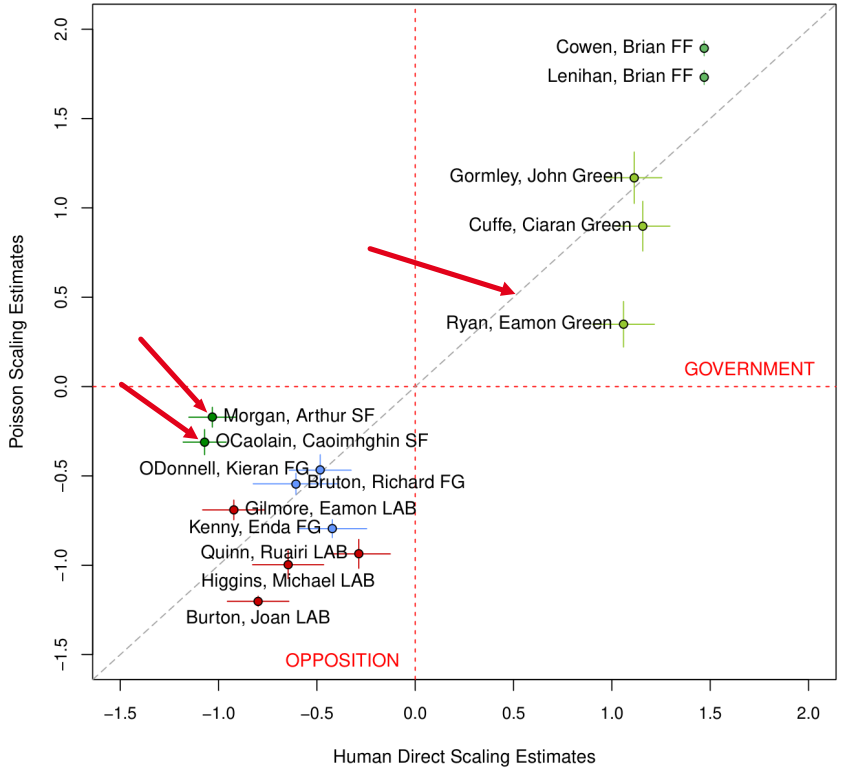


Figure 3: *Direct comparison of Human v. Wordfish Estimates.*

Figure 3 · Sinn Féin and the gray agreement line

Study 6

- Many positions broadly agree.
- **Sinn Féin:** strong opposition looks too moderate in Wordfish.
- **A second political distinction complicates one dimension.**

Lowe & Benoit (2013), Political Analysis, §§5–6. Figure 3 in the author manuscript, p. 32. Dimensionality background: Benoit (2018), p. 63.

Reading budget speeches without party labels

Study 6 · Reading exercise

Text A

> We have sought to do all of this in a manner that is fair and that protects the most vulnerable.

Text B

> This is a jobless and a joyless budget.

Discuss: which passage supports the budget? What else would you read before assigning a position?

Verbatim excerpts from `quanteda.textmodels::data_corpus_irishbudget2010`. Speaker identities follow on the next slide. Corpus documentation. Related reading example: Benoit (2018), pp. 64–66.

Budget support and speaking roles

Study 6 · Reading exercise

> We have sought to do all of this in a manner that is fair and that protects the most vulnerable.

Brian Lenihan (supports the budget) · Fianna Fáil · Minister for Finance

> This is a jobless and a joyless budget.

Richard Bruton (opposes the budget) · Fine Gael · opposition finance spokesperson

Interpretation: budget support, policy preferences and government responsibility can overlap in these speeches.

Irish 2010 budget corpus, Lenihan sentence 7 and Bruton sentence 6 under quanteda 4.3.1 sentence segmentation. Context: Lowe & Benoit (2013), §3. Related role example: Benoit (2018), pp. 64–66.

Gormley: one speaker, two roles

Study 6 · Lecture example

Benoit's fuller-speech example: Green Party leader and government minister.

> "As leader of the Green Party …"

The speaker sets out his party's budget position.

> "I will now comment on some specific aspects of my Department's Estimate."

The speaker turns to departmental business.

Discuss: which passages belong in a measure of budget support?

Benoit (2018), pp. 64–66; short excerpts from p. 65. The lecture's text scope differs from our classroom corpus.

Gormley: suggested answer

Study 6 · Lecture example

Target: support for or opposition to the budget.

- **Keep:** passages that express support for or criticism of the budget.
- **Leave out:** purely administrative details with no position on the budget.
- **Why:** those details can change the vocabulary and shift the score.

Decide from the content of each passage.

Reference: Benoit (2018), pp. 64–66.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

- Compare speakers **within each debate** first.
- Estimate a Wordfish position for each speaker in that debate.

- Connect debate-specific positions through legislators who appear across debates.

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

- **Compare speakers within each debate first.**
- Estimate a Wordfish position for each speaker in that debate.

- Connect debate-specific positions through legislators who appear across debates.

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

- Compare speakers **within each debate** first.
- **Estimate a Wordfish position for each speaker in that debate.**

- Connect debate-specific positions through legislators who appear across debates.

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

- Compare speakers **within each debate** first.
- Estimate a Wordfish position for each speaker in that debate.

- **Connect debate-specific positions through legislators who appear across debates.**

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

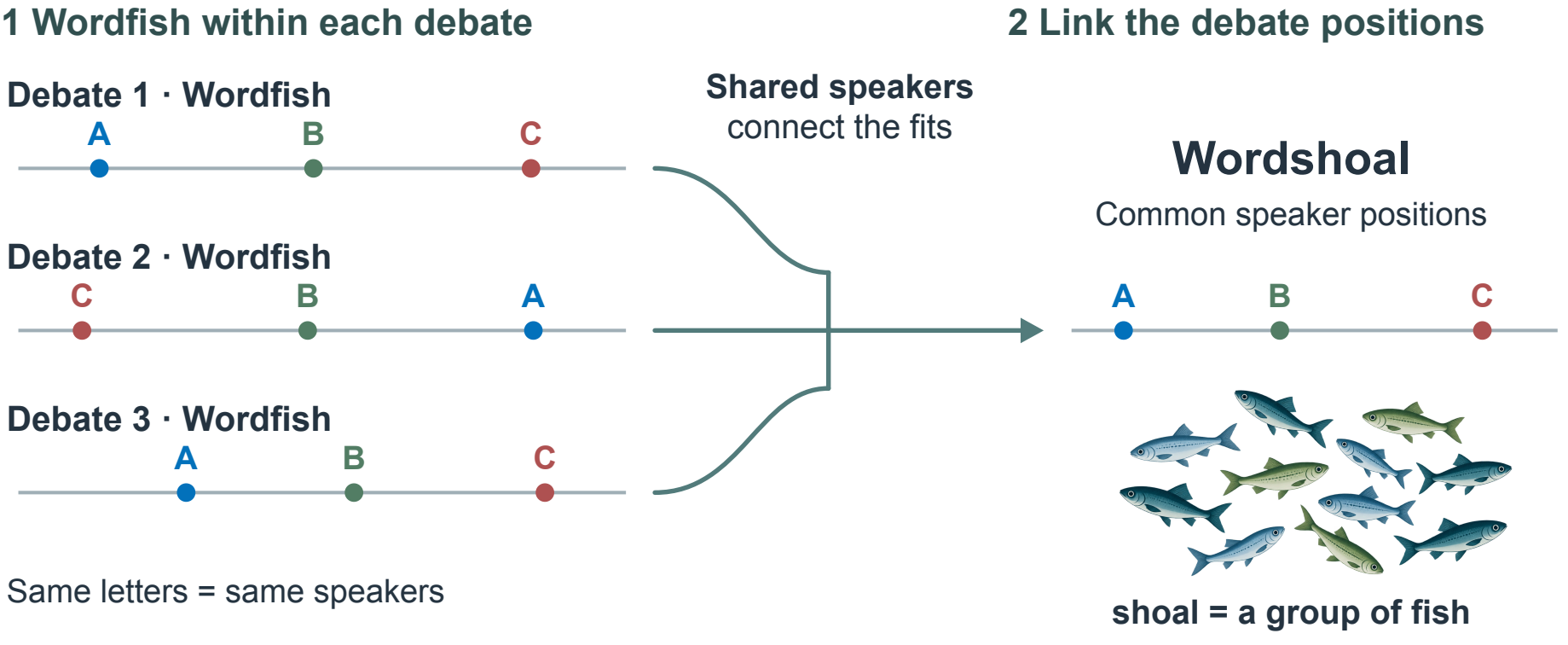
- Compare speakers **within each debate** first.
- Estimate a Wordfish position for each speaker in that debate.

- Connect debate-specific positions through legislators who appear across debates.

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: a shoal of Wordfish



Course schematic · A, B and C are speakers who recur across debates.

One Wordfish fit per debate; shared speakers link the results.

Lauderdale & Herzog (2016), Political Analysis, p. 374 n. 1 (name), pp. 377–378 (two stages). Course diagram; fish illustration generated with OpenAI.

Wordshoal: assumptions across debates

- **Within debates: word-use differences provide information about relative positions.**
- **Across debates:** some position differences reflect a common dimension.

- **Overlap:** repeated participants connect the debates.
- **Participation:** selective speaking limits what observed positions represent.

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: assumptions across debates

- **Within debates:** word-use differences provide information about relative positions.
- **Across debates: some position differences reflect a common dimension.**

- **Overlap:** repeated participants connect the debates.
- **Participation:** selective speaking limits what observed positions represent.

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: assumptions across debates

- **Within debates:** word-use differences provide information about relative positions.
- **Across debates:** some position differences reflect a common dimension.

- **Overlap: repeated participants connect the debates.**
- **Participation:** selective speaking limits what observed positions represent.

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: assumptions across debates

- **Within debates:** word-use differences provide information about relative positions.
- **Across debates:** some position differences reflect a common dimension.

- **Overlap:** repeated participants connect the debates.
- **Participation: selective speaking limits what observed positions represent.**

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: assumptions across debates

- **Within debates:** word-use differences provide information about relative positions.
- **Across debates:** some position differences reflect a common dimension.

- **Overlap:** repeated participants connect the debates.
- **Participation:** selective speaking limits what observed positions represent.

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Lauderdale and Herzog: legislative speech

Study 3

Measuring Political Positions from Legislative Speech

Benjamin E. Lauderdale
Department of Methodology, London School of Economics, Houghton Street, London WC2A 2AE, UK
e-mail: b.e.lauderdale@lse.ac.uk (corresponding author)

Alexander Herzog
School of Computing, Clemson University, Clemson, SC 29634, USA
e-mail: aherzog@clemson.edu

Benjamin E. Lauderdale and Alexander Herzog

Political Analysis, 2016

- **Question:** how can debates yield legislator-specific positions?
- **Ireland:** 29th and 30th Dáil.
- **United States:** 104th–113th Senates.

Lauderdale & Herzog (2016), "Measuring Political Positions from Legislative Speech", §§3–5. Lecture example: Benoit (2018), pp. 47–50.

Government and opposition in the Irish Dáil

Study 3

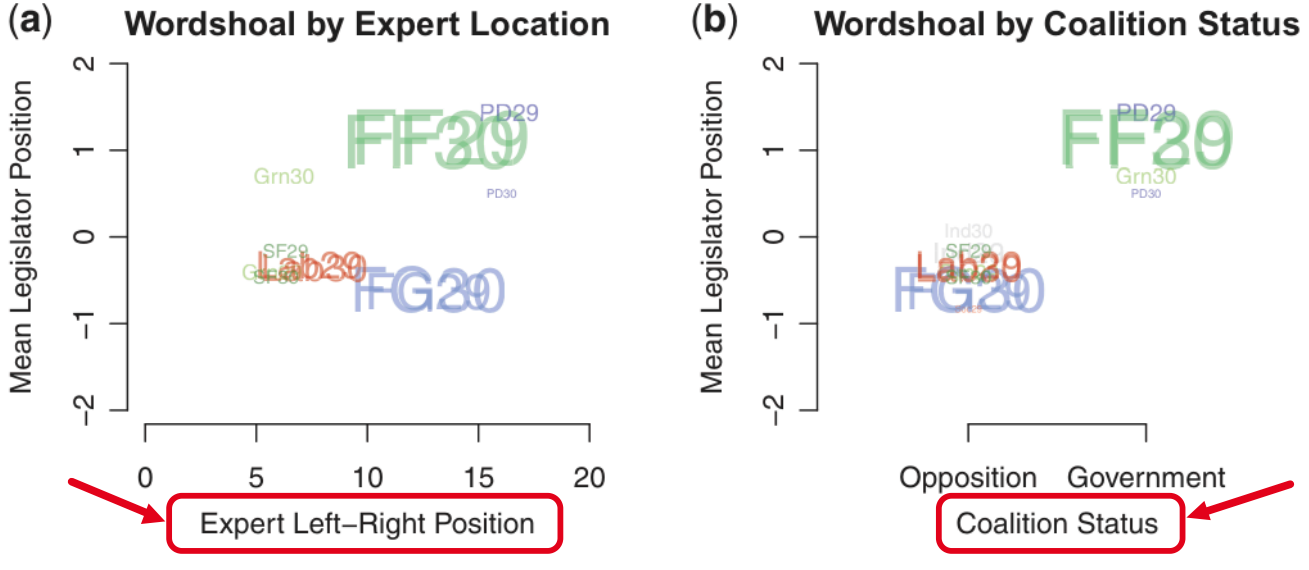


Figure 1 · panels a and b

- **Compare two possible interpretations.**
- The main dimension follows **coalition status.**
- Left–right needs its own evidence.

Lauderdale & Herzog (2016), Political Analysis, Figure 1(a–b), advance article p. 8. Original panels cropped from the full figure. Lecture example: Benoit (2018), p. 50.

Government and opposition in the Irish Dáil

Study 3

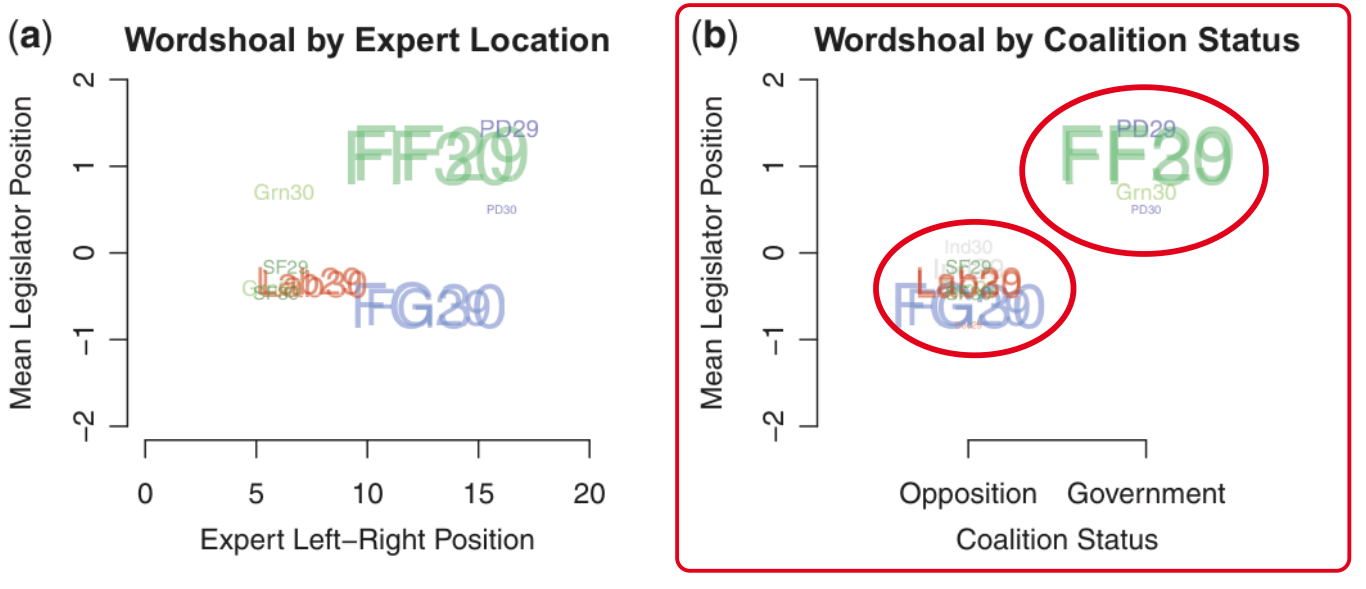


Figure 1 · panels a and b

- Compare two possible interpretations.
- **The main dimension follows coalition status.**
- Left–right needs its own evidence.

Lauderdale & Herzog (2016), Political Analysis, Figure 1(a–b), advance article p. 8. Original panels cropped from the full figure. Lecture example: Benoit (2018), p. 50.

Government and opposition in the Irish Dáil

Study 3

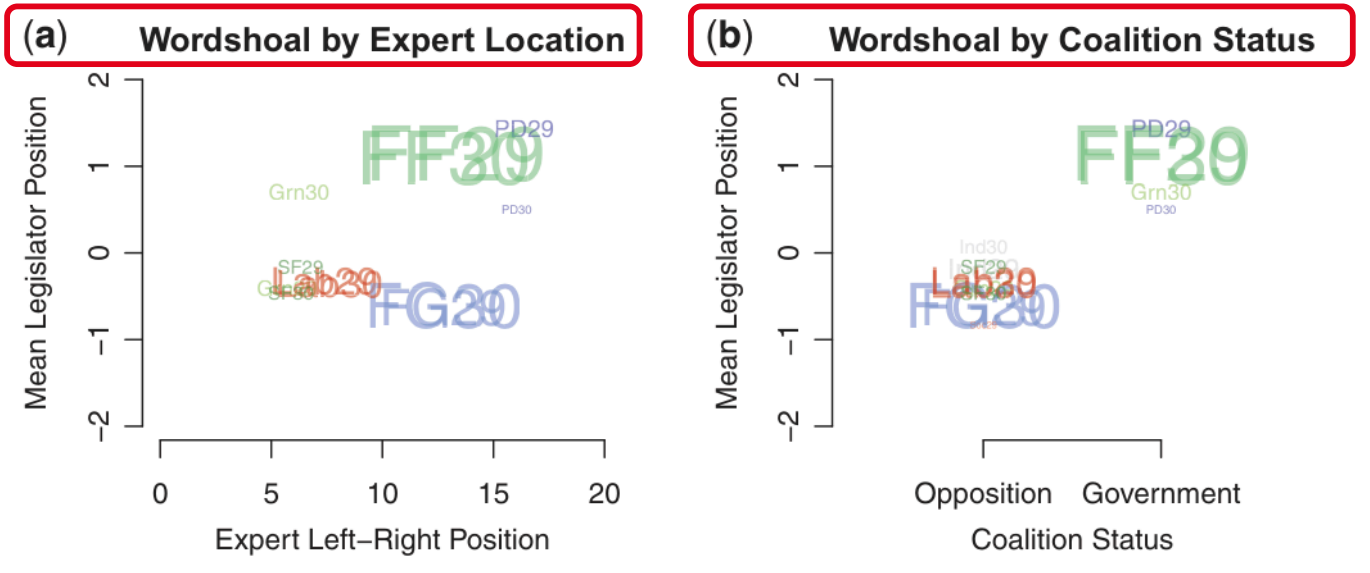


Figure 1 · panels a and b

- Compare two possible interpretations.
- The main dimension follows **coalition status.**
- **Left–right needs its own evidence.**

Lauderdale & Herzog (2016), Political Analysis, Figure 1(a–b), advance article p. 8. Original panels cropped from the full figure. Lecture example: Benoit (2018), p. 50.

Wordshoal versus Wordfish: coalition status

Study 3 · 30th Dáil

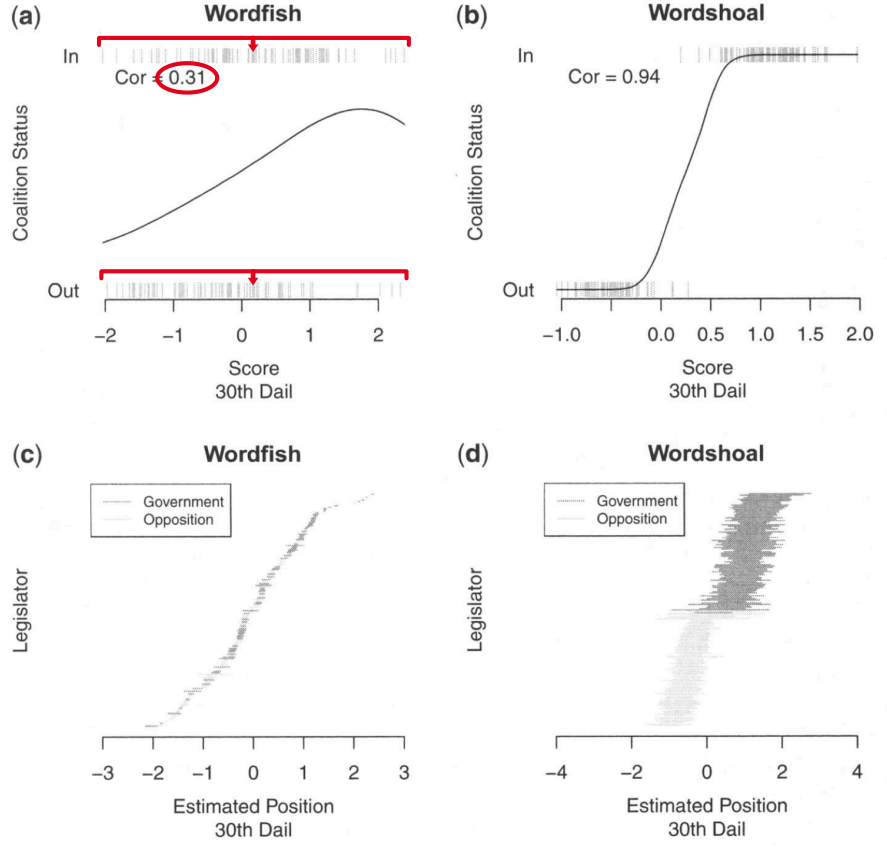


Figure 2 · individual legislators · panels a–d · red overlays

Individual legislators

- **Wordfish: r = 0.31. Government and opposition scores overlap.**
- **Wordshoal: r = 0.94.** Much clearer separation in this session.
- Bottom panels show positions and **95% intervals**.

r is the correlation with coalition status.

Lauderdale & Herzog (2016), Political Analysis, Figure 2, p. 382; interpretation, p. 383. Screenshot supplied by the instructor.

Wordshoal versus Wordfish: coalition status

Study 3 · 30th Dáil

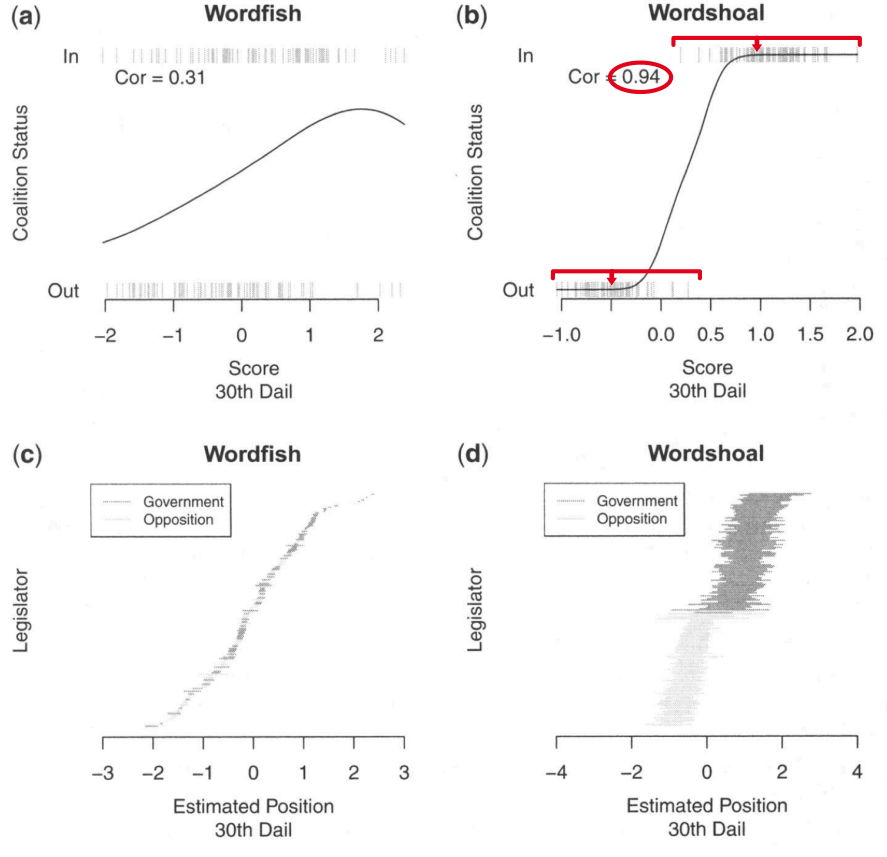


Figure 2 · individual legislators · panels a–d · red overlays

Individual legislators

- **Wordfish: r = 0.31.** Government and opposition scores overlap.
- **Wordshoal: r = 0.94. Much clearer separation in this session.**
- Bottom panels show positions and **95% intervals**.

r is the correlation with coalition status.

Lauderdale & Herzog (2016), Political Analysis, Figure 2, p. 382; interpretation, p. 383. Screenshot supplied by the instructor.

Debate structure and research scope

Study 3

Speaker	Budget	Health	Foreign policy
A	Speaks	Speaks	—
B	Speaks	—	Speaks
C	—	Speaks	Speaks

Teaching example: shared speakers link different debates.

A single budget debate provides no cross-debate structure. A disconnected set of speakers requires additional identifying information.

Illustrative participation table created for this course. Method connection: Lauderdale & Herzog (2016), §3. Lecture context: Benoit (2018), pp. 47–49.

LSS: seed words and a defined concept

Situation: we can name a concept's poles but have few labelled documents.

- Provide words that express each end of the intended dimension.
- Learn semantic relationships from a relevant corpus.
- Use proximity to the seeds to score words, then documents.

Hostility (+1): *enemy, foe, hostile*.
Friendliness (−1): *ally, friend, peaceful*.

Watanabe (2021), "Latent Semantic Scaling". Seed examples from Trubowitz & Watanabe (2021), ISQ. Plural forms and other seeds omitted here.

LSS: assumptions about meaning and context

- **Seeds:** words express the intended poles in this domain.
- **Semantics:** co-occurrence patterns contain useful information about that distinction.
- **Aggregation:** word scores provide a meaningful summary at the document level.
- **Target:** polarity relates to the actor or issue we intend to measure.

Discuss: could a negative news story describe an ally sympathetically?

Watanabe (2021), CMM. LSX: seed selection and target-specific example. Discussion is a course application.

Trubowitz and Watanabe: geopolitical threat

Study 4



The Geopolitical Threat Index: A Text-Based Computational Approach to Identifying Foreign Threats

PETER TRUBOWITZ
London School of Economics and Political Science, UK

AND

KOHEI WATANABE
Waseda University, Japan

Peter Trubowitz and Kohei Watanabe

International Studies Quarterly, 2021

- **Texts:** New York Times summaries, 1861–2017.
- **Models:** Newsmap and LSS.
- **Target:** reported hostile foreign military behavior.

Trubowitz & Watanabe (2021), "The Geopolitical Threat Index". LSS method: Watanabe (2021). Image: published advance article from LSE Research Online.

Hostility words and the threat index

Study 4

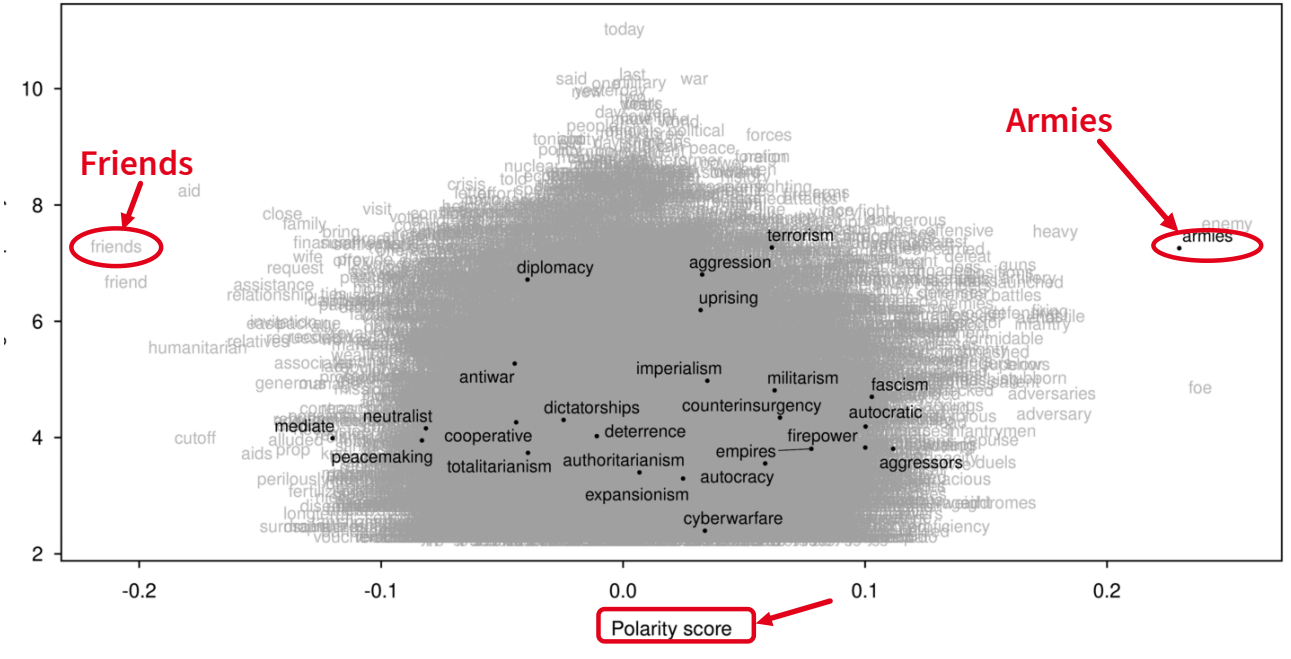


Figure 2 · learned word polarity

➜ **Seeds specify hostility.**

- Countries and relevant stories require separate identification.
- GTI aggregates the share of hostile military stories.

Trubowitz & Watanabe (2021), ISQ, Figure 2 and "Geopolitical Threat Index." GTI is a proportion, distinct from a mean polarity score.

Hostility words and the threat index

Study 4

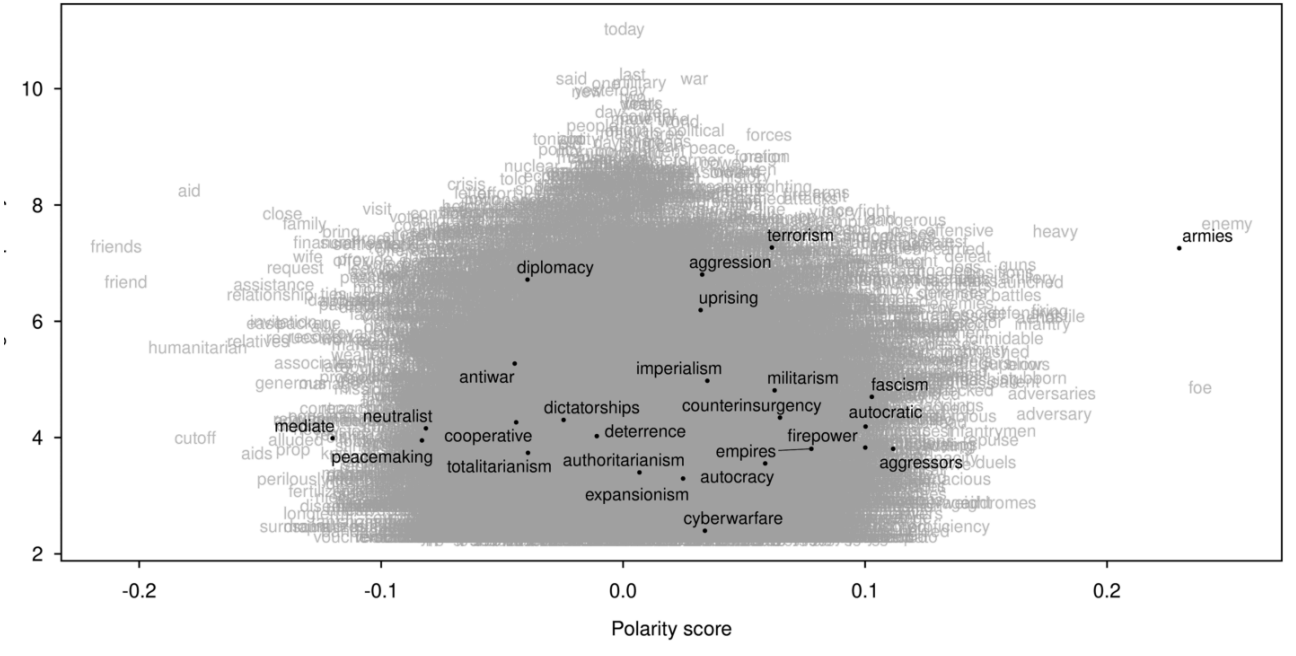


Figure 2 · learned word polarity

- Seeds specify **hostility**.

➜ **Countries and relevant stories require separate identification.**

- GTI aggregates the share of hostile military stories.

Trubowitz & Watanabe (2021), ISQ, Figure 2 and "Geopolitical Threat Index." GTI is a proportion, distinct from a mean polarity score.

Hostility words and the threat index

Study 4

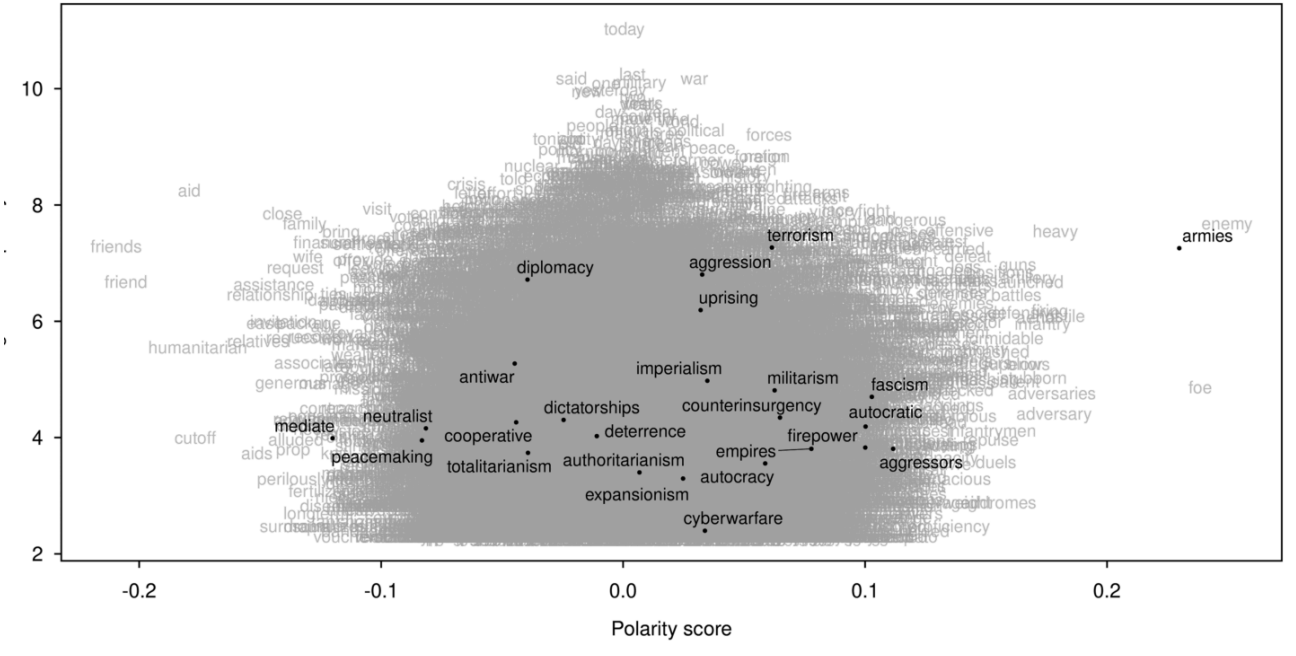


Figure 2 · learned word polarity

- Seeds specify **hostility**.
- Countries and relevant stories require separate identification.
- ➜ **GTI aggregates the share of hostile military stories.**

Trubowitz & Watanabe (2021), ISQ, Figure 2 and "Geopolitical Threat Index." GTI is a proportion, distinct from a mean polarity score.

Validation depends on the level of analysis

Study 4

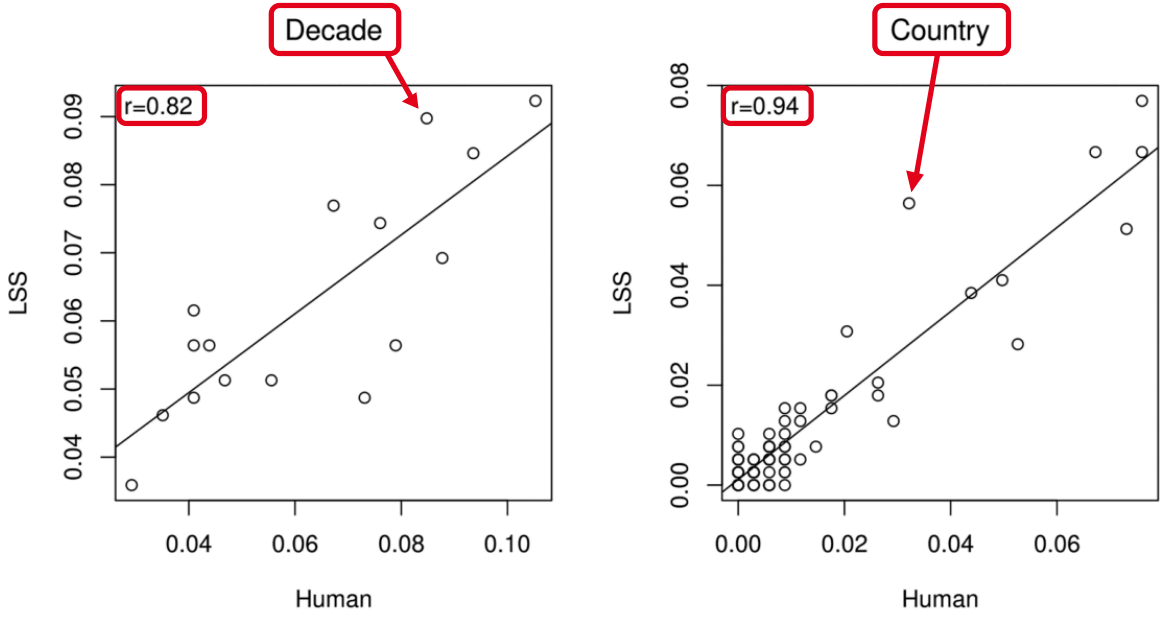


Figure 4 · aggregate hostile-story proportions

➜ **By decade: r = 0.82.**

➜ **By country: r = 0.94.**

- Individual summaries: about **59% agreement**.

Check the same unit your claim is about.

Reference: Trubowitz & Watanabe (2021), ISQ, Fig. 4; fn. 18.

Validation depends on the level of analysis

Study 4

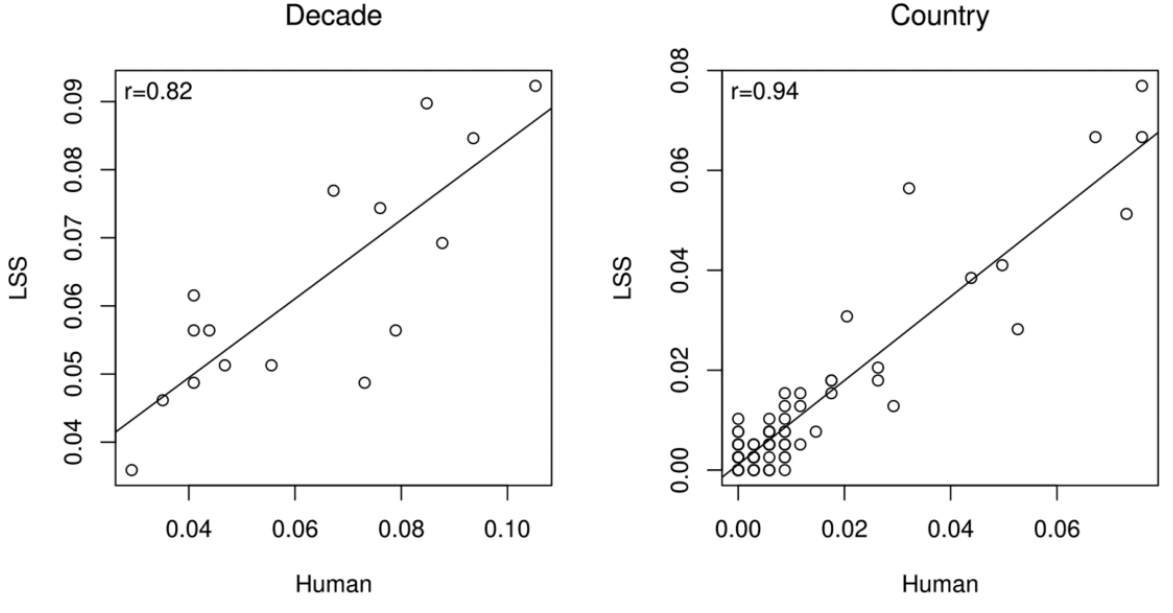


Figure 4 · aggregate hostile-story proportions

- By decade: **r = 0.82.**
- By country: **r = 0.94.**

➜ **Individual summaries: about 59% agreement.**

Check the same unit your claim is about.

Reference: Trubowitz & Watanabe (2021), ISQ, Fig. 4; fn. 18.

Validation depends on the level of analysis

Study 4

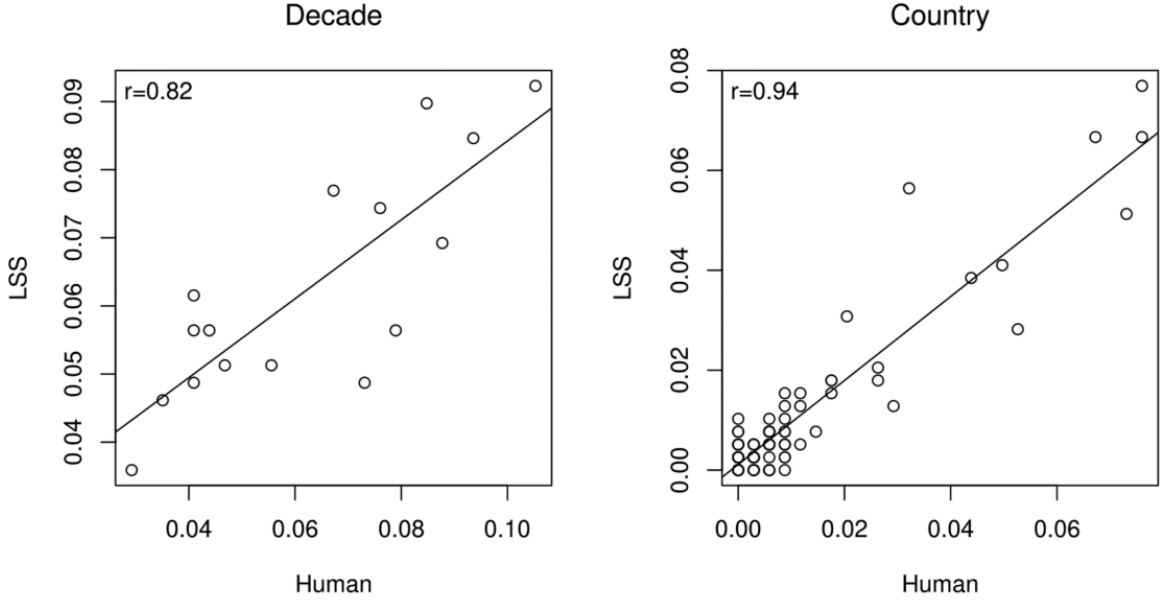


Figure 4 · aggregate hostile-story proportions

- By decade: **r = 0.82.**
- By country: **r = 0.94.**
- Individual summaries: about **59% agreement.**

➜ **Check the same unit your claim is about.**

Reference: Trubowitz & Watanabe (2021), ISQ, Fig. 4; fn. 18.

Supervision, units and outputs

Studies 1–4

Method	Guidance supplied	Structure	Main output
Wordscores	Reference document scores	Comparable reference and new texts	New document positions
Wordfish	No position labels	One count matrix	Relative document positions
Wordshoal	No position labels	Speakers across debates	Common speaker positions
LSS	Seed words and polarity	Corpus semantic relationships	Word and document polarity

Next: connect speakers across debates with Wordshoal.

Laver et al. (2003) · Slapin & Proksch (2008) · Lauderdale & Herzog (2016) · Watanabe (2021). Lecture context: Benoit (2018), pp. 4, 34–35, 47–49.

Wordshoal: linking debate positions

First stage: estimate a Wordfish position z_{ij} within each debate j.

$$z_{ij} = a_j + b_j \theta_i + \varepsilon_{ij}$$

Quantity	Meaning
θ_i	**Speaker's common position**
b_j	Direction and strength of a debate's connection
Missing z_{ij}	The legislator did not speak in that debate

Debate scales can connect with different signs and strengths.

Teaching notation for Lauderdale & Herzog (2016), §3.2. The original second-stage model also specifies speaker precision and priors. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: linking debate positions

First stage: estimate a Wordfish position z_{ij} within each debate j.

$$z_{ij} = a_j + b_j \theta_i + \varepsilon_{ij}$$

Quantity	Meaning
θ_i	Speaker's common position
b_j	**Direction and strength of a debate's connection**
Missing z_{ij}	The legislator did not speak in that debate

Debate scales can connect with different signs and strengths.

Teaching notation for Lauderdale & Herzog (2016), §3.2. The original second-stage model also specifies speaker precision and priors. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: linking debate positions

First stage: estimate a Wordfish position $z_{i\,j}$ within each debate j.

$$z_{ij} = a_j + b_j \theta_i + \varepsilon_{ij}$$

Quantity	Meaning
θ_i	Speaker's common position
b_j	Direction and strength of a debate's connection
Missing $z_{i\,j}$	**The legislator did not speak in that debate**

Debate scales can connect with different signs and strengths.

Teaching notation for Lauderdale & Herzog (2016), §3.2. The original second-stage model also specifies speaker precision and priors. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: align debate scales first

Constructed example: two debates encode the same ordering with opposite signs.

Speaker	Debate A position	Debate B position	Naive average
A	−1	+1	0
B	0	0	0
C	+1	−1	0

A common model must align debate locations, scales and directions before interpreting shared positions.

Discuss: why does a negative debate loading not necessarily indicate an absence of common structure?

Benoit (2018), pp. 47–49; Lauderdale & Herzog (2016). This is not output from a Wordshoal fit.

LSS: semantic information and aggregation

Step	Information used
Define poles	**Concept-specific seed words and polarity**
Estimate semantic relationships	Word co-occurrence, typically across sentences
Score words	Similarity to the seed words in a reduced semantic space
Score documents	Frequency-weighted average of word scores

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: semantic information and aggregation

Step	Information used
Define poles	Concept-specific seed words and polarity
Estimate semantic relationships	**Word co-occurrence, typically across sentences**
Score words	Similarity to the seed words in a reduced semantic space
Score documents	Frequency-weighted average of word scores

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: semantic information and aggregation

Step	Information used
Define poles	Concept-specific seed words and polarity
Estimate semantic relationships	Word co-occurrence, typically across sentences
Score words	**Similarity to the seed words in a reduced semantic space**
Score documents	Frequency-weighted average of word scores

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: semantic information and aggregation

Step	Information used
Define poles	Concept-specific seed words and polarity
Estimate semantic relationships	Word co-occurrence, typically across sentences
Score words	Similarity to the seed words in a reduced semantic space
Score documents	**Frequency-weighted average of word scores**

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: semantic information and aggregation

Step	Information used
Define poles	Concept-specific seed words and polarity
Estimate semantic relationships	Word co-occurrence, typically across sentences
Score words	Similarity to the seed words in a reduced semantic space
Score documents	Frequency-weighted average of word scores

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: from vectors to word polarity

Use singular value decomposition (**SVD**) to reduce sentence–word counts to **k dimensions**.

$$g_j = (1 / |S|) \sum_{s \in S} \cos(v_s, v_j)\, p_s$$

Quantity	Meaning
v_j	Word vector learned from sentence contexts
p_s	Researcher-assigned polarity of seed s, such as −1 or +1
g_j	Word polarity: similarity weighted by the seed polarities

Document scores then average word polarities using token frequencies.

Watanabe (2021), "Latent Semantic Scaling," Eqs. 1–2. Representation background: Benoit (2018), pp. 54, 68–69.

From W6 evaluation to W7 validation

Question	W6 classification practice	W7 scaling
What is evaluated?	Predicted categories	Positions on a stated dimension
What is the evidence?	Reserved human labels; F1	External measures; rank agreement
What do we read?	Errors and coding boundaries	Surprising positions and their language

Hjorth et al. test convergence across measures. This differs from k-fold cross-validation.

Next: do Wordscores and Wordfish agree with external measures of party positions?

Course synthesis connecting W6, Hjorth et al. (2015) and Lowe & Benoit (2013). Validation framework: Benoit (2018), p. 59.

Hjorth et al.: validating party positions

Study 5

Computers, coders, and voters: Comparing automated methods for estimating party positions

Frederik Hjorth[1],Robert Klemmensen[2],Sara Hobolt[3], Martin Ejnar Hansen[4],Peter Kurrild-Klitgaard[5]

Frederik Hjorth, Robert Klemmensen, Sara Hobolt, Martin Ejnar Hansen and Peter Kurrild-Klitgaard

Research & Politics, 2015

- **Question:** do text scales agree with other measures of party positions?
- **Comparison:** Wordscores and Wordfish.
- **Scope:** 254 manifestos across 33 elections.

Hjorth et al. (2015), "Computers, Coders, and Voters", pp. 1–2. Original article heading, CC BY-NC 3.0.

Two countries, three validation benchmarks

Study 5

Country	Elections	Manifestos	Mean words
Denmark, 1945–2007	24	212	1,232
Germany, 1980–2009	9	42	10,306

- **Experts:** published assessments of party positions.
- **Voters:** mean left–right self-placement of each party's supporters.
- **CMP RILE:** a left–right index from human-coded manifestos.

Assumption to examine: do these benchmarks address the same political dimension?

Hjorth et al. (2015), pp. 2–4 and Table 1. Mean word counts rounded. CMP = Comparative Manifesto Project.

Validation differs across the two countries

Study 5

Original Table 3: mean Spearman rank correlations across elections.

Country	Method	CMP	Experts	Voters
Denmark	Wordscores	0.7	0.8	0.8
Denmark	Wordfish	0.2	0.2	0.1
Germany	Wordscores	0.8	0.8	0.8
Germany	Wordfish	0.8	0.7	0.7

Rank agreement supports an ordering; it does not establish meaningful distances.

Discuss: why might Wordfish agree with the benchmarks in one country more than the other?

Hjorth et al. (2015), Table 3, p. 7. Published correlations transcribed; estimation times omitted. These are not classification accuracies.

Why did Wordfish do better in Germany?

Study 5

Explanation	What the authors report
Text length: checked	Randomly shorten German manifestos to a Danish-like average. About half of Germany's Wordfish advantage disappears.
Political vocabulary: inferred	Shortened German texts still do better. The authors link the remaining advantage to more ideologically polarized language.

Evidence limit: this is the authors' rough attribution. Vocabulary was not separately tested.

Hjorth et al. (2015), "Difference in cross-validity," p. 4, and discussion, pp. 5–6. Approximate attribution on p. 4. Detailed shortening results are in the authors' online appendix.

Anchors and independent validation

Evidence	What it does	What it leaves to check
Wordscores references	Define the learned scale	Performance on other texts
Wordfish `dir`	Choose an orientation	Meaning of the dimension
LSS seeds	Specify a semantic direction	Concept and target in context
Independent judgments	Check an intended interpretation	Agreement, coverage and scope

Party separation and agreement between models provide partial evidence.

Course synthesis of Lowe & Benoit (2013) and the model documentation. Judgments used to tune a measure should be separated from its final evaluation. Validation framework: Benoit (2018), p. 59.

Robustness and uncertainty

Sensitivity: replace references or seeds, change the text scope, and compare reasonable preprocessing choices.

Uncertainty: distinguish model intervals, human disagreement and variation across specifications.

Evaluation: reserve judgments that did not guide those choices.

Discuss: which choice would you vary first, and what evidence would you reserve?

Lowe & Benoit (2013), uncertainty and model assumptions. Watanabe, seed-word evaluation. Evaluation design is a course synthesis. Lecture context: Benoit (2018), pp. 60–66.

A validation plan for your research abstract

1. **Concept:** what does a high score mean?
2. **Unit:** which text, speaker or period receives the score?
3. **Design:** what information and structure does the method require?
4. **Assumption:** which condition is most likely to fail in your data?
5. **Evidence:** what independent observation could reveal that failure?

Research title and abstract: due **October 23, 2026, before 23:59**.

Course schedule. Deadline follows the W6 announcement. Validation questions synthesize the W7 research cases. Validation framework: Benoit (2018), p. 59.

R tools for text scaling

Task	Package	Main interface
Prepare texts and counts	`quanteda`	Corpus, tokens and DFM
Reference-based scaling	`quanteda.textmodels`	`textmodel_wordscores()`
Latent count scaling	`quanteda.textmodels`	`textmodel_wordfish()`
Plot estimated positions	`quanteda.textplots`	`textplot_scale1d()`
Cross-debate scaling	`wordshoal`	`textmodel_wordshoal()`
Semantic scaling	`LSX`	`textmodel_lss()`

R practice: Wordscores and Wordfish on one budget debate.

quanteda · quanteda.textmodels · wordshoal · LSX.

R practice: one debate, two models

Target: support for the 2010 Irish budget.

- Fit Wordscores and Wordfish to the same fourteen speeches.
- Compare rankings and read the largest disagreement.
- Replace one reference and assess sensitivity.

Work in pairs. Run S01–S08b, including lettered substeps, in one R session.
Lab HTML · Lab ZIP.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 59, 64–66.

S01: Packages and the Irish corpus

Copy and run S01.

```r
# S01: Packages and the built-in corpus
# install.packages(c("quanteda", "quanteda.textmodels",
#                    "quanteda.textplots"))
library(quanteda)
library(quanteda.textmodels)
library(quanteda.textplots)
quanteda_options(threads = 1)
options(width = 72, digits = 4)
corp_irish <- data_corpus_irishbudget2010
c(quanteda = as.character(packageVersion("quanteda")),
  textmodels = as.character(packageVersion("quanteda.textmodels")),
  textplots = as.character(packageVersion("quanteda.textplots")))
ndoc(corp_irish)
```

The corpus is included in quanteda.textmodels. Keep this R session open for all later steps.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP.

S01 output: packages and documents

R console output

```
   quanteda textmodels  textplots
    "4.3.1"   "0.9.10"   "0.96.1"
[1] 14
```

Explanation

- The built-in corpus contains fourteen speeches, each kept as one document.
- Version strings identify the installed software, helping us trace differences between runs.
- `[1]` marks the first printed value; `14` is the document count.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP.

S02a: Speakers and document IDs

Copy and run S02a.

```r
# S02a: Speakers and stable document IDs
meta <- data.frame(
  doc_id = paste0("ie_", docvars(corp_irish, "number")),
  speaker = docvars(corp_irish, "name"),
  party = docvars(corp_irish, "party"))
docnames(corp_irish) <- meta$doc_id
print(meta, row.names = FALSE)
```

Each row is one speech. The stable IDs keep texts, references and model results aligned.

S02a output: the fourteen speakers

R console output

```
  doc_id  speaker party
  ie_01  Lenihan    FF
  ie_02   Bruton    FG
  ie_03   Burton   LAB
  ie_04   Morgan    SF
  ie_05    Cowen    FF
  ie_06    Kenny    FG
  ie_07 ODonnell    FG
  ie_08  Gilmore   LAB
  ie_09  Higgins   LAB
  ie_10    Quinn   LAB
  ie_11  Gormley Green
  ie_12     Ryan Green
  ie_13    Cuffe Green
  ie_14 OCaolain    SF
```

Explanation

- IDs come from speech numbers, so each score can be matched to the correct text after sorting.
- `speaker` and `party` describe the speeches; neither model uses the party column to fit positions.

S02b: Tokenize a working copy

Copy and run S02b.

```r
# S02b: Tokenize a working copy
toks_irish <- tokens(corp_irish, remove_punct = TRUE,
  remove_symbols = TRUE, remove_numbers = TRUE)
toks_irish <- tokens_tolower(toks_irish)
head(as.list(toks_irish)[[1]], 12)
```

Remove punctuation, symbols and numbers, then lowercase the working tokens. Keep the original corpus for reading.

S02b output: words from the first speech

R console output

```
[1] "when"          "i"             "presented"     "the"
[5] "supplementary" "budget"        "to"            "this"
[9] "house"         "last"          "april"         "i"
```

Explanation

- The twelve words stay in speech order because this previews tokens, before counting them.
- `[1]`, `[5]`, and `[9]` mark token positions, not frequencies.
- Lowercase words and missing punctuation follow our cleaning choices; the later matrix also removes order.

S02c: Build the count matrix

Copy and run S02c.

```
# S02c: Build the common count matrix
counts <- dfm(toks_irish)
dim(counts)
any(ntoken(counts) == 0)
topfeatures(counts, 6)
```

Both methods start from these token counts. No TF-IDF weighting is applied.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

S02c output: counts and common words

R console output

```
[1]   14 4885
[1] FALSE
 the   to   of  and   in    a
3598 1633 1537 1359 1231 1013
```

Explanation

- `14  4885` reflects fourteen speeches and their combined vocabulary; cells count word occurrences.
- `FALSE` confirms cleaning left words in every speech.
- We kept stopwords, so `the` remains common: 3,598 occurrences alone do not distinguish political positions.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

S03: Define the Wordscores references

Copy and run S03.

```
# S03: Define the Wordscores references
ref_scores <- rep(NA_real_, ndoc(counts))
names(ref_scores) <- docnames(counts)
ref_ids <- c("ie_01", "ie_02")
ref_scores[ref_ids] <- c(1, -1)
print(data.frame(speaker = meta$speaker[match(ref_ids, meta$doc_id)],
  reference = ref_scores[ref_ids]), row.names = FALSE)
```

Assign +1 to Lenihan and −1 to Bruton for budget support. Leave the other twelve positions unknown.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S03 output: the reference design

R console output

```
  speaker reference
Lenihan         1
 Bruton        -1
```

Explanation

- We assign Lenihan +1 because he defends the budget, and Bruton –1 because he criticizes it.
- These inputs set the intended direction; the other twelve positions stay NA because we have not supplied them.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S04a: Fit Wordscores and read word scores

Copy and run S04a.

```
# S04a: Fit Wordscores and inspect word scores
ws <- textmodel_wordscores(counts, y = ref_scores, smooth = 0)
probe_words <- c("budget", "government", "tax",
                 "jobs", "fair", "recovery")
print(data.frame(word = probe_words,
  score = round(coef(ws)[probe_words], 3)), row.names = FALSE)
```

With smooth = 0, words must occur in the reference vocabulary to receive a score.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 9–13.

S04a output: six learned word scores

R console output

```
      word  score
    budget -0.384
government  0.140
       tax  0.665
      jobs  0.379
      fair -0.229
  recovery  0.403
```

Explanation

- `tax` scores `+0.665` because it is proportionally more frequent in Lenihan's positive reference.
- `fair` scores `−0.229` because it is proportionally more frequent in Bruton's negative reference.
- Scores follow reference usage, so ordinary positive or negative sentiment does not determine their signs.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 9–13.

S04b: Check the scoring vocabulary

Copy and run S04b.

```
# S04b: Align the scoring vocabulary and inspect coverage
ws_counts <- dfm_match(counts, features = names(coef(ws)))
ws_coverage <- ntoken(ws_counts) / ntoken(counts)
dim(ws_counts)
summary(ws_coverage)
```

Match the DFM to words scored by the references. Coverage is the share of each speech's processed tokens that remain.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14, 19.

S04b output: vocabulary and token coverage

R console output

```
[1]   14 2131
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.836   0.850   0.874   0.886   0.887   1.000
```

Explanation

- Only 2,131 of 4,885 features occur in the references and receive scores; all fourteen speeches remain.
- Coverage is 83.6%–100%, with median 87.4%, because speeches use different amounts of scored vocabulary.
- Repeated occurrences count toward coverage; this measures retained language, not prediction accuracy.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14, 19.

S04c: Predict raw Wordscores

Copy and run S04c.

```r
# S04c: Predict raw Wordscores for every speech
ws_pred <- predict(ws, newdata = ws_counts, rescaling = "none")
results <- meta
results$reference <- ref_scores
results$ws_raw <- as.numeric(ws_pred)
results$coverage <- ws_coverage
print(data.frame(speaker = results$speaker, party = results$party,
  reference = results$reference,
  ws_raw = round(results$ws_raw, 3)), row.names = FALSE)
```

The reference column records assigned inputs. ws_raw records predictions, including reference-text rescoring.

S04c output: reference and new speech scores

R console output

speaker	party	reference	ws_raw
Lenihan	FF	1	0.339
Bruton	FG	−1	−0.339
Burton	LAB	NA	−0.001
Morgan	SF	NA	0.023
Cowen	FF	NA	0.180
Kenny	FG	NA	−0.020
ODonnell	FG	NA	−0.029
Gilmore	LAB	NA	−0.012
Higgins	LAB	NA	−0.040
Quinn	LAB	NA	−0.023
Gormley	Green	NA	0.159
Ryan	Green	NA	−0.012
Cuffe	Green	NA	0.003
OCaolain	SF	NA	0.026

Explanation

- Assigned `+1` and `−1` teach word scores; rescoring averages those scores, yielding Lenihan `+0.339` and Bruton `−0.339`.
- Shared vocabulary pulls raw averages toward the centre.
- `NA` marks twelve speeches whose reference positions were not supplied; `ws_raw` contains their predictions.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14–15.

S04d: Plot raw Wordscores positions

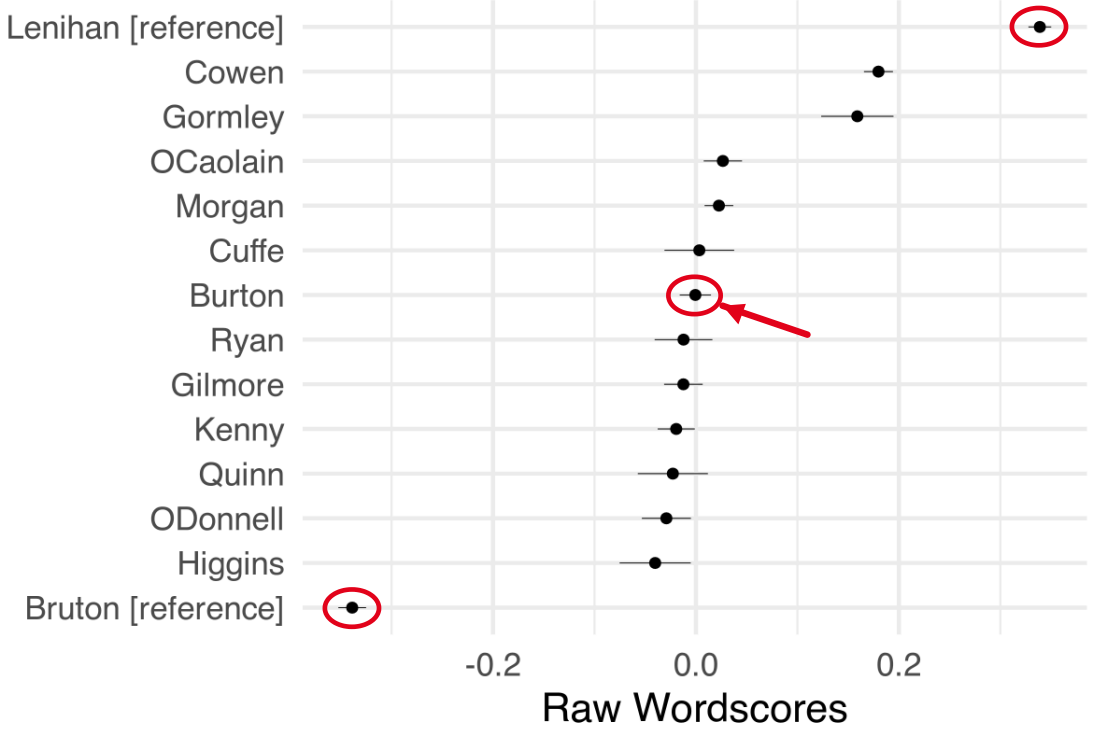
Copy and run S04d.

```
# S04d: Plot raw Wordscores with model-based intervals
ws_plot <- predict(ws, newdata = ws_counts,
  se.fit = TRUE, rescaling = "none")
ws_labels <- paste0(meta$speaker,
  ifelse(is.na(ref_scores), "", " [reference]"))
textplot_scale1d(ws_plot, doclabels = ws_labels, sort = TRUE) +
  ggplot2::labs(x = NULL, y = "Raw Wordscores") +
  ggplot2::theme_minimal(base_size = 24)
```

Plot all fourteen raw predictions. The two reference speeches are labelled; their dots are rescored values, not assigned endpoints.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Plot: quanteda example. Concepts: Benoit (2018), pp. 14–15.

S04d output: Wordscores positions and intervals



Explanation

- Reference dots near `+0.339` and `−0.339` show rescored word averages, rather than assigned endpoints.
- Burton near zero reflects a word-score average; it does not establish neutral budget support.
- Bars show approximate 95% uncertainty under the scoring method; reference-choice uncertainty is excluded.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Plot: quanteda example. Concepts: Benoit (2018), pp. 14–15.

S05a: Fit Wordfish and set its direction

Copy and run S05a.

```r
# S05a: Fit Wordfish and fix its orientation
wf_dir <- match(c("ie_02", "ie_01"), docnames(counts))
set.seed(7197)
wf <- textmodel_wordfish(counts, dir = wf_dir,
  dispersion = "poisson")
data.frame(speaker = meta$speaker[wf_dir],
  theta = round(wf$theta[wf_dir], 3))
```

Estimate all positions jointly. dir asks for Bruton below Lenihan, using document indices found from stable IDs.

S05a output: the orientation check

R console output

```
  speaker  theta
1  Bruton -0.585
2 Lenihan  1.775
```

Explanation

- Bruton (−0.585) is below Lenihan (1.775) because `dir` chooses that ordering.
- The positions are estimated from counts, not fixed at −1/+1. Direction alone does not establish budget-support meaning.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 39–41.

S05b: Read Wordfish positions

Copy and run S05b.

```r
# S05b: Inspect Wordfish positions and model standard errors
results$wf_theta <- wf$theta
results$wf_se <- wf$se.theta
print(data.frame(speaker = results$speaker, party = results$party,
  theta = round(results$wf_theta, 3),
  se = round(results$wf_se, 3)), row.names = FALSE)
```

theta gives relative positions. se reports uncertainty under this fitted model and its assumptions.

S05b output: positions and standard errors

R console output

```
   speaker party  theta    se
   Lenihan    FF  1.775 0.021
    Bruton    FG -0.585 0.029
    Burton   LAB -1.150 0.016
    Morgan    SF -0.093 0.029
     Cowen    FF  1.794 0.024
     Kenny    FG -0.789 0.026
  ODonnell    FG -0.493 0.044
   Gilmore   LAB -0.587 0.030
   Higgins   LAB -0.979 0.041
     Quinn   LAB -0.921 0.042
   Gormley Green  1.122 0.075
      Ryan Green  0.212 0.065
     Cuffe Green  0.793 0.073
  OCaolain    SF -0.098 0.039
```

Explanation

- Burton has `theta = −1.150` and `se = 0.016`; Gormley's `se = 0.075` is larger.
- Burton's smaller standard error gives a narrower model interval; it does not prove the scale measures budget support.

S05c: Read Wordfish word parameters

Copy and run S05c.

```
# S05c: Inspect word discrimination and baseline frequency
word_id <- match(probe_words, wf$features)
print(data.frame(word = probe_words,
  beta = round(wf$beta[word_id], 3),
  psi = round(wf$psi[word_id], 3)), row.names = FALSE)
```

beta links word use to position. psi represents the word's baseline frequency.

S05c output: discrimination and baseline frequency

R console output

```
      word  beta   psi
    budget 0.042 2.711
government 0.173 2.612
       tax 0.396 2.194
      jobs 0.505 1.842
      fair 0.191 0.860
  recovery 0.358 0.897
```

Explanation

- Budget's `psi` (2.711) describes baseline commonness; its small `beta` (0.042) means little change along theta.
- Jobs' `beta` (0.505) gives stronger log expected-count change per `theta` unit, with other effects fixed.
- `psi` is not raw frequency; positive `beta` is not positive sentiment.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

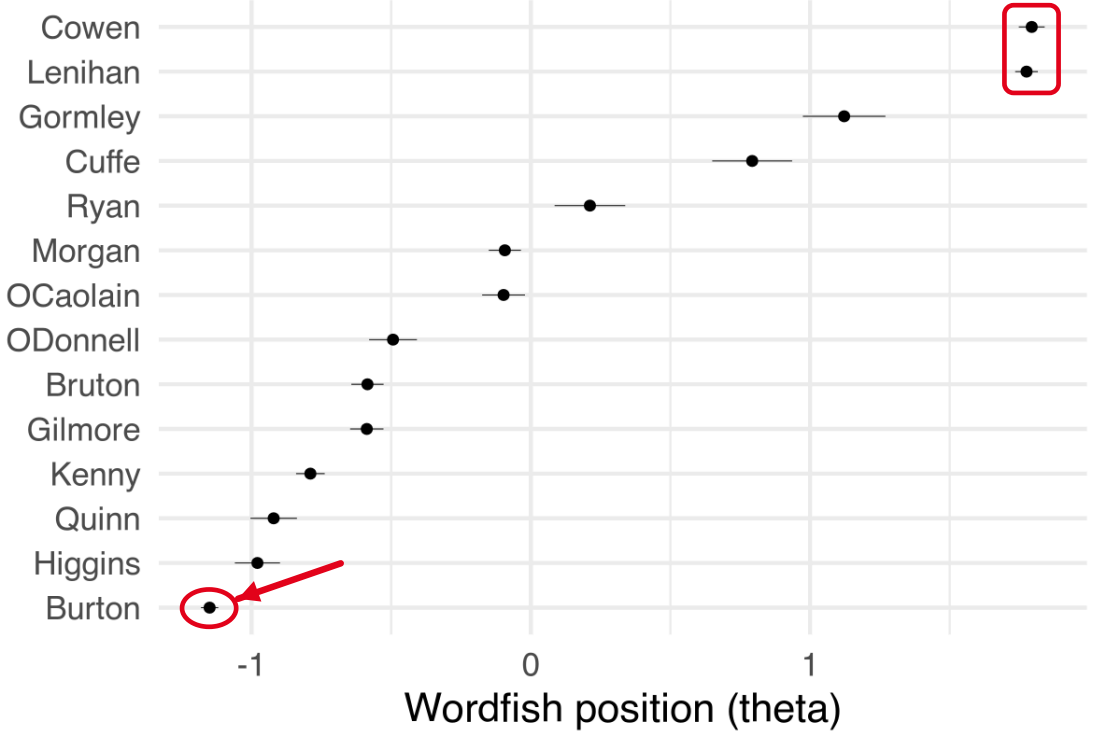
S05d: Plot Wordfish positions

Copy and run S05d.

```
# S05d: Plot Wordfish positions with model-based intervals
textplot_scale1d(wf, doclabels = meta$speaker, sort = TRUE) +
  ggplot2::labs(x = NULL, y = "Wordfish position (theta)") +
  ggplot2::theme_minimal(base_size = 24)
```

Plot the fourteen estimated positions before comparing models. The horizontal bars use the fitted model standard errors.

S05d output: Wordfish positions and intervals



Explanation

- Cowen (1.794) and Lenihan (1.775) lie to the right of Burton (−1.150) on the theta axis.
- Approximate 95% bars use `theta ± 1.96 × se`; smaller standard errors produce narrower intervals.
- Intervals describe model uncertainty; they do not establish budget-support meaning.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Plot: quanteda example. Concepts: Benoit (2018), pp. 59–61.

S06a: Compare the twelve new texts

Copy and run S06a.

```r
# S06a: Compare rankings among the twelve new texts
comparison <- results[is.na(results$reference), ]
comparison$ws_rank <- rank(comparison$ws_raw)
comparison$wf_rank <- rank(comparison$wf_theta)
comparison$gap <- abs(comparison$ws_rank - comparison$wf_rank)
comparison <- comparison[order(-comparison$gap, comparison$doc_id), ]
round(cor(comparison$ws_raw, comparison$wf_theta,
  method = "spearman"), 3)
print(head(comparison[c("speaker", "ws_rank", "wf_rank", "gap")], 6),
  row.names = FALSE)
```

Exclude both Wordscores references. Rank 1 is the lower end of each scale, after fixing a common orientation.

S06a output: ranking agreement and disagreements

R console output

```
[1] 0.734
  speaker ws_rank wf_rank gap
   Burton       7       1   6
 ODonnell       2       6   4
     Ryan       6       9   3
 OCaolain      10       7   3
    Cuffe       8      10   2
   Morgan       9       8   1
```

Explanation

- `ws_rank = 7` and `wf_rank = 1` give Burton an absolute `gap = 6`; rank 1 means lower.
- Spearman 0.734 summarizes twelve speeches: overall agreement can coexist with one large disagreement. It is not accuracy.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 59.

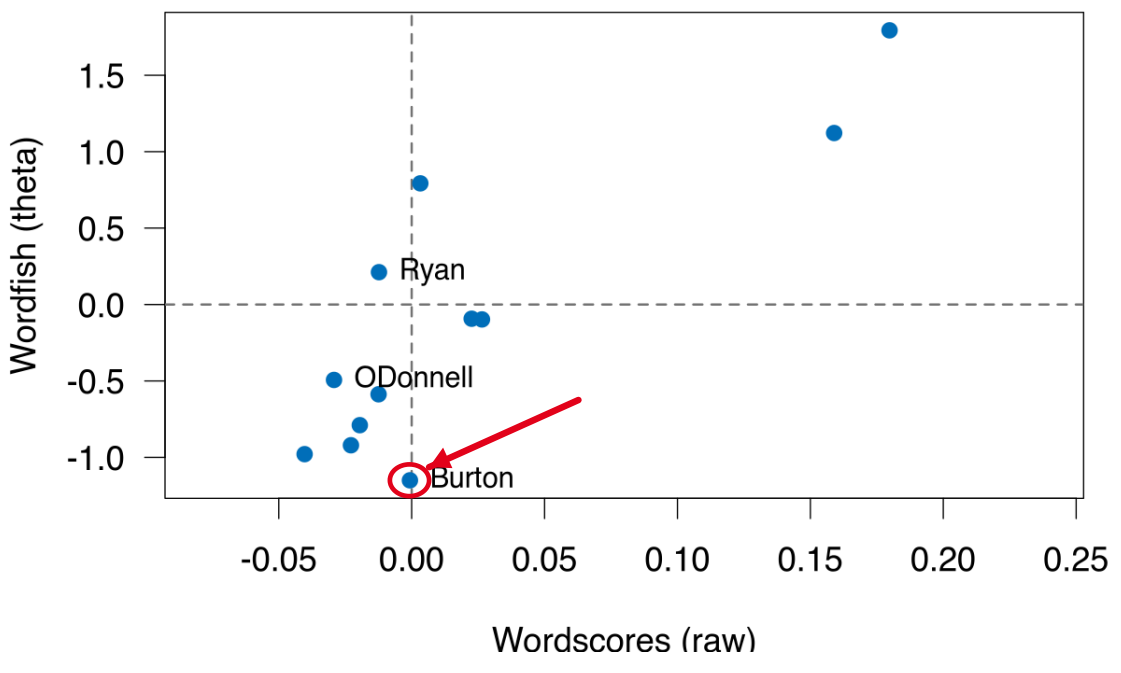
S06b: Plot the two scales

Copy and run S06b.

```r
# S06b: Plot the new texts on both scales
plot(comparison$ws_raw, comparison$wf_theta,
  type = "n", xlim = c(-0.08, 0.24),
  xlab = "Wordscores (raw)", ylab = "Wordfish (theta)")
abline(h = 0, v = 0, col = "grey45", lty = 2, lwd = 1.8)
points(comparison$ws_raw, comparison$wf_theta,
  pch = 19, col = "#0271bb")
text(comparison$ws_raw[1:3], comparison$wf_theta[1:3],
  labels = comparison$speaker[1:3], pos = 4, cex = 0.85)
```

Each point is one of the twelve new texts. Labels identify the three largest ranking disagreements.

S06b output: two scales for the new texts



Explanation

- Burton is near zero in Wordscores but low in Wordfish: ranks 7 versus 1.
- Wordscores uses two reference speeches; Wordfish fits all speeches' word-count patterns.
- The axes have different units. Read her argument before treating near-zero as neutral.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 41.

S07a: Read a ranking disagreement

Copy and run S07a.

```r
# S07a: Identify and read the largest ranking disagreement
read_id <- comparison$doc_id[1]
print(comparison[1, c("doc_id", "speaker", "ws_rank", "wf_rank")],
  row.names = FALSE)
read_sentences <- as.character(corpus_reshape(
  corp_irish[read_id], to = "sentences"))
cat(strwrap(read_sentences[1], width = 66), sep = "\n")
```

Use the largest ranking gap to select a speech, then return to its original wording.

S07a output: the selected speech and its opening

R console output

```
  doc_id speaker ws_rank wf_rank
  ie_03  Burton       7       1
The other day, Goldman Sachs circulated a note about the Irish
budget which referred to the remarkable degree of consensus in
Ireland about the necessity for budget adjustments.
```

Explanation

- Burton was selected because her model ranks differ most: Wordscores 7, Wordfish 1.
- The consensus claim comes from a Goldman Sachs note she reports, so it does not establish her approval.

S07b: Read the surrounding argument

Copy and run S07b.

```
# S07b: Read more of the same speech
# In this corpus, the selected speech is Burton's. Read sentences 5–
  7.
cat(strwrap(paste(read_sentences[5:7], collapse = " "),
  width = 66), sep = "\n")
# For the full speech in your console:
# cat(as.character(corp_irish[read_id]))
```

Read sentences 5–7 in Burton’s speech. The original corpus preserves their punctuation and wording.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 64–66.

S07b output: more of the same speech

R console output

```
One could never have imagined one's party giving such an
advantageous start to any Government. This could have been a
consensus budget as it was a classic "all jump together" moment,
an opportunity for burden sharing by everyone but the Minister
blew it. The budget is straight out of the Leona Helmsley school
of thought in which burden sharing, like taxation, is only for
the little people.
```

Explanation

- A `consensus budget` is something Burton says the Minister failed to achieve; she criticizes burden sharing.
- Words like consensus appear inside criticism, so a near-zero score need not mean moderate support.

S08a: Change one reference text

Copy and run S08a.

```r
# S08a: Replace the positive reference and refit Wordscores
ref_alt <- ref_scores
ref_alt[c("ie_01", "ie_05")] <- c(NA, 1)
ws_alt <- textmodel_wordscores(counts, y = ref_alt, smooth = 0)
ws_alt_counts <- dfm_match(counts, features = names(coef(ws_alt)))
ws_alt_pred <- predict(ws_alt, newdata = ws_alt_counts,
  rescaling = "none")
print(data.frame(speaker = meta$speaker[!is.na(ref_alt)],
  reference = ref_alt[!is.na(ref_alt)]), row.names = FALSE)
```

Replace Lenihan with Cowen at +1. Keep Bruton at −1 and use the same preprocessing and raw scoring rule.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08a output: the alternative reference design

R console output

```
  speaker reference
   Bruton        -1
    Cowen         1
```

Explanation

- `reference` lists assigned inputs: Cowen +1 replaces Lenihan +1; Bruton −1 stays.
- A different reference speech supplies different word shares, so word scores and scoring vocabulary can change despite identical endpoints.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08b: Compare a common set of new texts

Copy and run S08b.

```
# S08b: Compare ranks within the same eleven new texts
common_new <- is.na(ref_scores) & is.na(ref_alt)
sensitivity <- meta[common_new, c("doc_id", "speaker")]
sensitivity$before <- rank(ws_pred[common_new])
sensitivity$after <- rank(ws_alt_pred[common_new])
sensitivity$change <- sensitivity$after - sensitivity$before
print(sensitivity[c("speaker", "before", "after", "change")],
  row.names = FALSE)
```

Recalculate both rankings within the same eleven texts. Exclude Lenihan, Bruton and Cowen from this comparison.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08b output: sensitivity to the positive reference

R console output

speaker	before	after	change
Burton	7	5	−2
Morgan	9	8	−1
Kenny	4	2	−2
ODonnell	2	4	2
Gilmore	5	6	1
Higgins	1	1	0
Quinn	3	3	0
Gormley	11	11	0
Ryan	6	9	3
Cuffe	8	10	2
OCaolain	10	7	−3

Explanation

- Ryan: 6→9 (+3); O'Caolain: 10→7 (−3). These are the largest absolute changes.
- `change = after − before`, with both rankings recalculated among the same eleven speeches.
- Changing reference language changes learned word scores, so rankings can move without any policy change.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

Reporting a text-scaling analysis

Decision	What to report
Corpus and representation	Debate, speakers, preprocessing and counts
Wordscores	Reference texts, assigned values, coverage and raw/rescaled scores
Wordfish	Direction, model settings and the interpretation of theta
Assessment	Ranking disagreements, passages read and sensitivity

Return to Lowe & Benoit: what independent reader judgments would test our budget-support interpretation?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 59.

References: text scaling foundations

Benoit, Kenneth. 2018. *Scaling Text*. Quants 3: Quantitative Text Analysis, Trinity College Dublin, March 23. Lecture reference for the concept walkthrough, pp. 3–27, 34–50, 54–55, 59–70.

Laver, Michael, and Kenneth Benoit. 2002. "Locating TDs in Policy Spaces: Wordscoring Dáil Speeches." *Irish Political Studies* 17(1): 59–73. DOI.

Laver, Michael, Kenneth Benoit, and John Garry. 2003. "Extracting Policy Positions from Political Texts Using Words as Data." *American Political Science Review* 97(2): 311–331. DOI.

Slapin, Jonathan B., and Sven-Oliver Proksch. 2008. "A Scaling Model for Estimating Time-Series Party Positions from Texts." *American Journal of Political Science* 52(3): 705–722. DOI.

Lauderdale, Benjamin E., and Alexander Herzog. 2016. "Measuring Political Positions from Legislative Speech." *Political Analysis* 24(3): 374–394. DOI.

References: validation and semantic scaling

Hjorth, Frederik, Robert Klemmensen, Sara Hobolt, Martin Ejnar Hansen, and Peter Kurrild-Klitgaard. 2015. "Computers, Coders, and Voters: Comparing Automated Methods for Estimating Party Positions." *Research & Politics* 2(2): 1–9. DOI.

Lowe, Will, and Kenneth Benoit. 2013. "Validating Estimates of Latent Traits from Textual Data Using Human Judgment as a Benchmark." *Political Analysis* 21(3): 298–313. DOI.

Watanabe, Kohei. 2021. "Latent Semantic Scaling: A Semisupervised Text Analysis Technique for New Domains and Languages." *Communication Methods and Measures* 15(2): 81–102. DOI.

Trubowitz, Peter, and Kohei Watanabe. 2021. "The Geopolitical Threat Index: A Text-Based Computational Approach to Identifying Foreign Threats." *International Studies Quarterly* 65(3): 852–865. DOI.

Software and source materials

- quanteda and quanteda.textmodels: corpus preparation, Wordscores and Wordfish.
- wordshoal: cross-debate scaling and implementation documentation.
- LSX: semantic scaling and seed-word evaluation.
- Quanteda scaling tutorials: worked model examples.

Article images reproduce the original research evidence. The participation table and numerical Wordscores example are teaching illustrations. Speech excerpts retain their original wording.